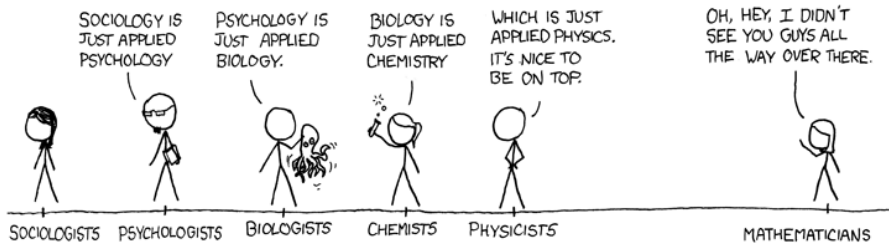




*Flujo de inferencia*

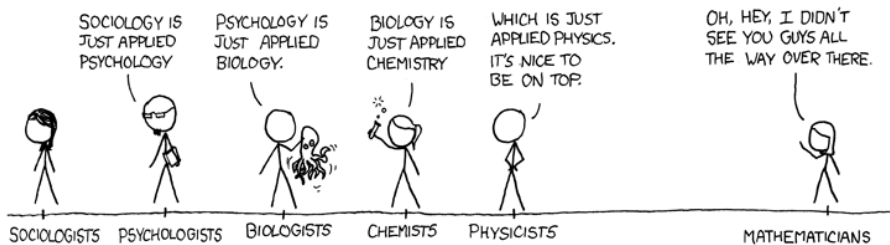
# Ciencias con datos

# Ciencias formales



## Ciencias con datos

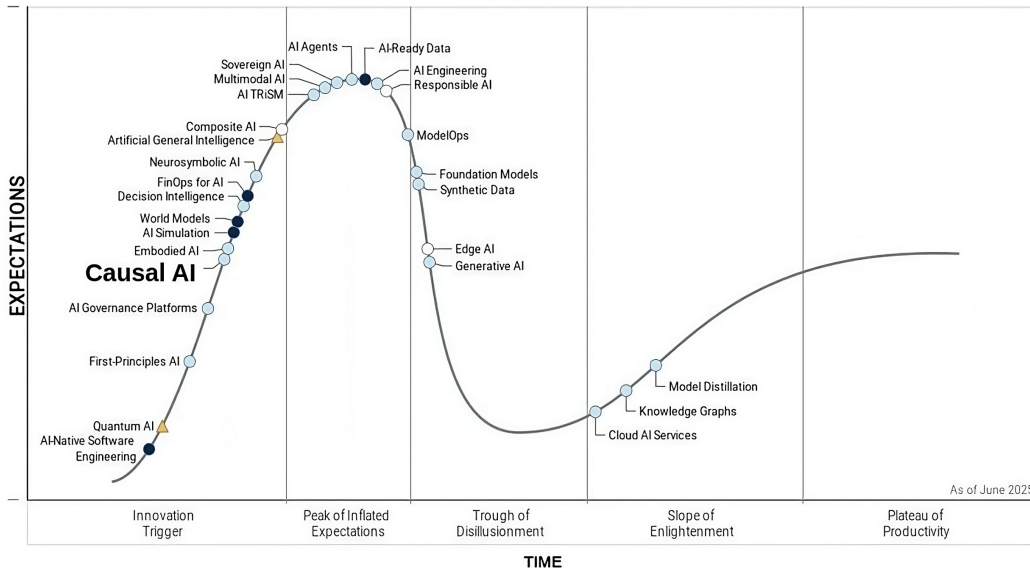
## Ciencias formales



Todas las ciencias con datos desarrollan  
**teorías causales** para comprender el mundo

# Industrias con datos

## Hype Cycle for Artificial Intelligence, 2025



# La ventaja de las teorías causales

$$\overbrace{-\log \lim_{T \rightarrow \infty} P(\text{Datos}_T | \text{Modelo Causal})^{1/T}}^{\text{Tasa de predicción en órdenes de magnitud}}$$

# La ventaja de las teorías causales

$$\begin{array}{c} \text{Tasa de predicción en} \\ \text{órdenes de magnitud} \\ \hline -\log \lim_{T \rightarrow \infty} P(\text{Datos}_T | \text{Modelo Causal})^{1/T} \\ \\ = \\ \sum_{c,s,r} \underbrace{P(c, s, r | \text{Realidad Causal})}_{\text{Probabilidad de que se genere el dato}} \cdot \underbrace{(-\log P(c, s, r | \text{Modelo Causal}))}_{\text{Información en órdenes de magnitud}} \\ \hline \text{Entropía cruzada} \end{array}$$

# La ventaja de las teorías causales

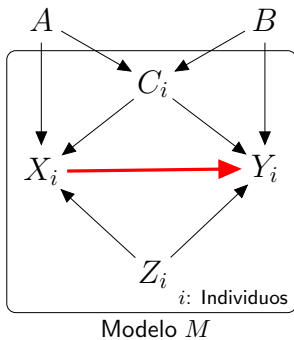
$$\begin{array}{c} \text{Tasa de predicción en} \\ \text{órdenes de magnitud} \\ \hline -\log \lim_{T \rightarrow \infty} P(\text{Datos}_T | \text{Modelo Causal})^{1/T} \\ \\ = \\ \sum_{c,s,r} \underbrace{P(c, s, r | \text{Realidad Causal})}_{\text{Probabilidad de que se genere el dato}} \cdot \underbrace{(-\log P(c, s, r | \text{Modelo Causal}))}_{\text{Información en órdenes de magnitud}} \\ \hline \text{Entropía cruzada} \end{array}$$

**No hay mejor predicción que la del modelo causal**  
que se corresponde con la realidad causal subyacente

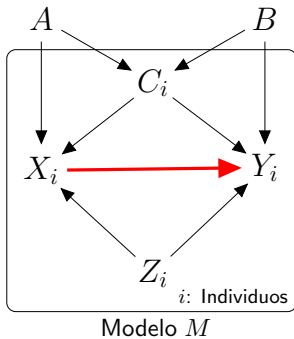
# Predecir el impacto de las acciones

a partir de intervenciones experimentales

# Predecir el impacto de las acciones a partir de intervenciones experimentales

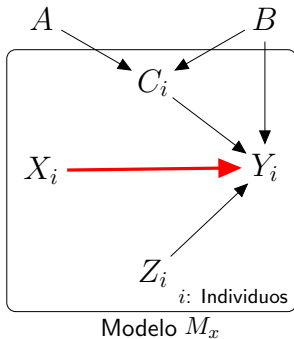


# Predecir el impacto de las acciones a partir de intervenciones experimentales



En los experimentos *aleatorizados* (y en las intervenciones determinísticas) el tratamiento se asigna independientemente de sus causas naturales.

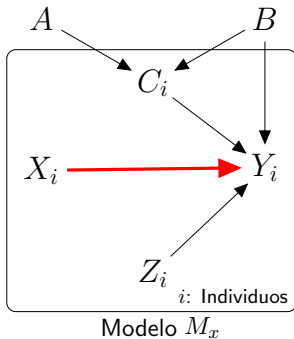
# Predecir el impacto de las acciones a partir de intervenciones experimentales



$\text{do}(X_i)$

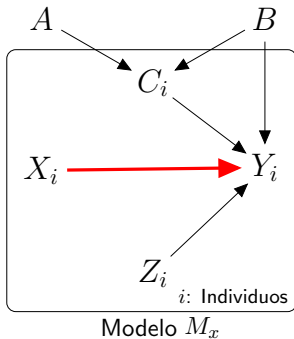
La intervención modifica el mecanismo causal de  $X$ :  $P(\text{do}(X_i = x)) = \mathbb{I}(X_i = x)$

# Predecir el impacto de las acciones a partir de intervenciones experimentales



$$\underbrace{P(y_i | \text{do}(x_i), M)}_{\text{El impacto de la intervención}} \underbrace{=}_{\text{es}} \underbrace{P(y_i | x_i, M_x)}_{\text{la distribución condicional en el modelo intervenido}}$$

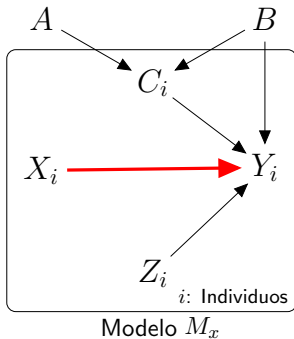
# Predecir el impacto de las acciones a partir de intervenciones experimentales



$$P(y_i | \text{do}(x_i)) = P_{M_x}(y_i | x_i)$$

otra notación común

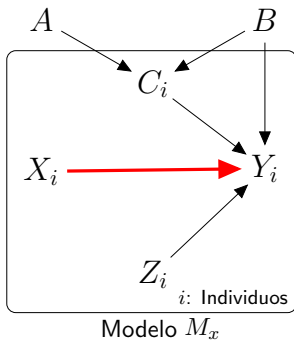
# Predecir el impacto de las acciones a partir de intervenciones experimentales



$$P(y_i|\text{do}(x_i)) = P_{M_x}(y_i|x_i)$$

el operador  $\text{do}()$  modifica la realidad causal subyacente

# Predecir el impacto de las acciones a partir de intervenciones experimentales



$$P(y_i|\text{do}(x_i)) = P_{M_x}(y_i|x_i)$$

en el modelo intervenido toda la asociación es causal (no hay asociación espuria)

# Predecir el impacto de las acciones a partir de datos observados

$$P(Y|X)$$



# Predecir el impacto de las acciones

a partir de datos observados

$$P(Y|X)$$

$\neq$

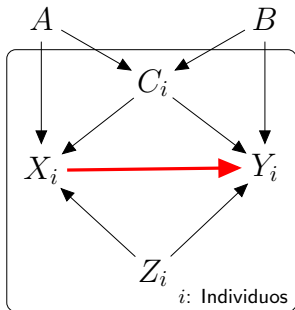
$$P(Y|\text{do}(X))$$

# Predecir el impacto de las acciones a partir de datos observados

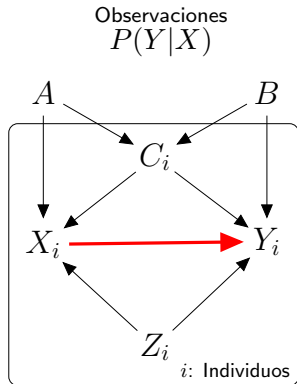
Observaciones  
 $P(Y|X)$

$\neq$

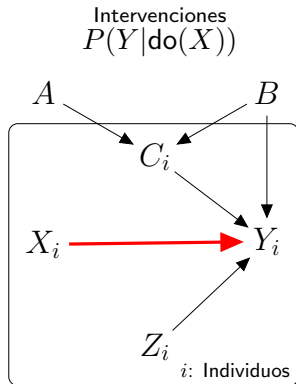
$P(Y|\text{do}(X))$



# Predecir el impacto de las acciones a partir de datos observados

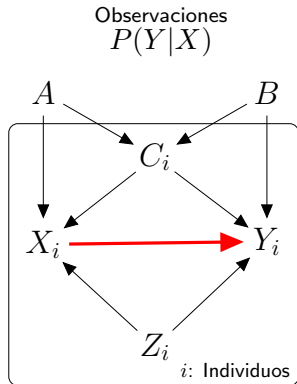


$\neq$

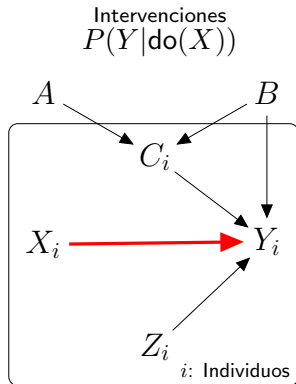


# Predecir el impacto de las acciones a partir de datos observados

## Entrenamos In-sample

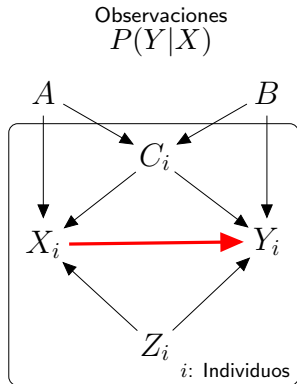


$\neq$



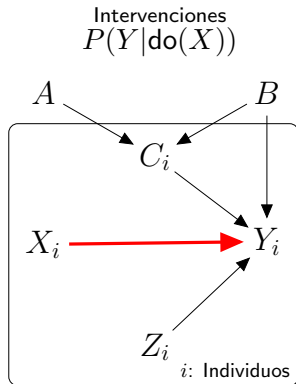
# Predecir el impacto de las acciones a partir de datos observados

**Entrenamos**  
In-sample



$\neq$

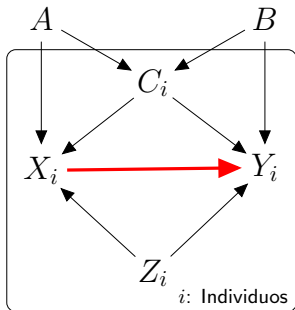
**Predecimos**  
Out-of-sample



# Predecir el impacto de las acciones a partir de datos observados

**Entrenamos**  
In-sample

Observaciones  
 $P(Y|X)$



$\neq$

**Predecimos**  
Out-of-sample

Intervenciones  
 $P(Y|\text{do}(X))$

La inferencia  $P_{M_x}(y|x)$  circula  
únicamente por los caminos causales

# Predecir el impacto de las acciones

a partir de datos observados

**Entrenamos**

In-sample

Observaciones

$$P(Y|X)$$

$\neq$

**Predecimos**

Out-of-sample

Intervenciones

$$P(Y|\text{do}(X))$$

La inferencia  $P(y|x)$  circula  
también por caminos no causales

La inferencia  $P_{M_x}(y|x)$  circula  
únicamente por los caminos causales

# Predecir el impacto de las acciones

a partir de datos observados

**Entrenamos**

In-sample

Observaciones  
 $P(Y|X)$

$\neq$

**Predecimos**

Out-of-sample

Intervenciones  
 $P(Y|\text{do}(X))$

La inferencia  $P(y|x)$  circula  
también por caminos no causales

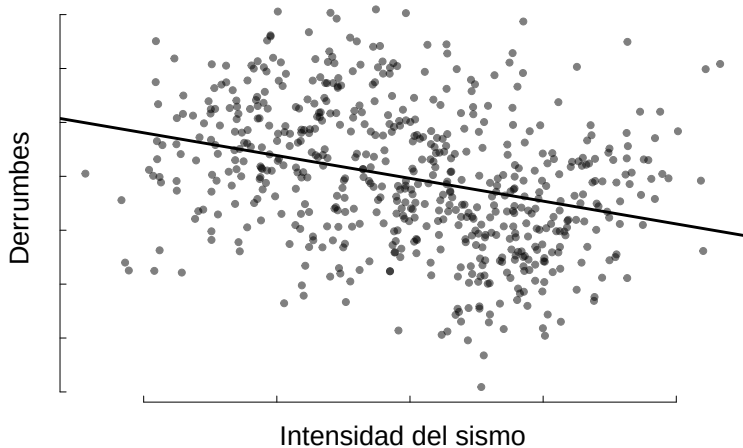
La inferencia  $P_{M_x}(y|x)$  circula  
únicamente por los caminos causales

**¿Podemos hacer el salto?**

Hacer inferencia sin experimentos que prediga el impacto causal

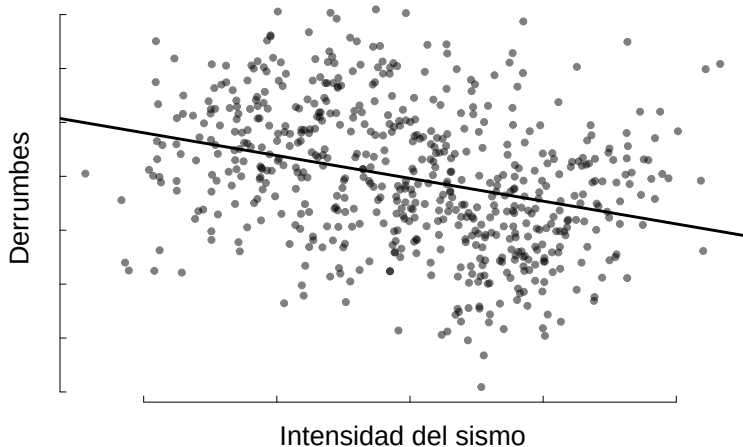
# Predecir el impacto de las acciones

a partir de datos observados



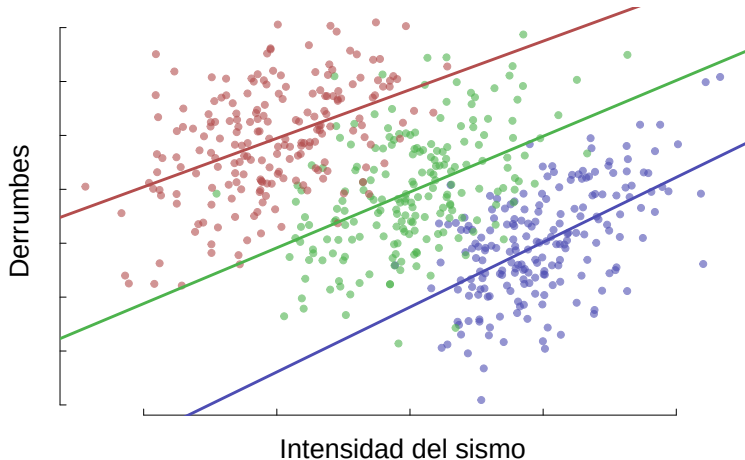
# Predecir el impacto de las acciones a partir de datos observados

$$P(\text{Derrumbes}|\text{Intensidad del sismo})$$



# Predecir el impacto de las acciones a partir de datos observados

$$P(\text{Derrumbes}|\text{Intensidad del sismo, Ciudad})$$



¿Cómo **predecir** el **impacto causal**  
a partir de **datos observados sin intervenciones**?

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

¿El Tratamiento es efectivo para mejorar el estado del paciente?

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$ | $E_0 = 1$ |  |
|---------|-----------|-----------|--|
| $T = 0$ |           |           |  |
| $T = 1$ |           |           |  |
|         |           |           |  |

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$                   | $E_0 = 1$ |
|---------|-----------------------------|-----------|
| $T = 0$ | $P(E_1 = 1 T = 0, E_0 = 0)$ |           |
| $T = 1$ | $P(E_1 = 1 T = 1, E_0 = 0)$ |           |
|         |                             |           |

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$                                                         | $E_0 = 1$ |
|---------|-------------------------------------------------------------------|-----------|
| $T = 0$ | $P(E_1 = 1 T = 0, E_0 = 0)$                                       |           |
| $T = 1$ | $P(E_1 = 1 T = 1, E_0 = 0)$                                       |           |
|         | $P(E_1=1 \mathbf{T=0}, E_0=0)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=0)$ |           |

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$                                                         | $E_0 = 1$                   |  |
|---------|-------------------------------------------------------------------|-----------------------------|--|
| $T = 0$ | $P(E_1 = 1 T = 0, E_0 = 0)$                                       | $P(E_1 = 1 T = 0, E_0 = 1)$ |  |
| $T = 1$ | $P(E_1 = 1 T = 1, E_0 = 0)$                                       | $P(E_1 = 1 T = 1, E_0 = 1)$ |  |
|         | $P(E_1=1 \mathbf{T=0}, E_0=0)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=0)$ |                             |  |

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$                                                         | $E_0 = 1$                                                         |  |
|---------|-------------------------------------------------------------------|-------------------------------------------------------------------|--|
| $T = 0$ | $P(E_1 = 1 T = 0, E_0 = 0)$                                       | $P(E_1 = 1 T = 0, E_0 = 1)$                                       |  |
| $T = 1$ | $P(E_1 = 1 T = 1, E_0 = 0)$                                       | $P(E_1 = 1 T = 1, E_0 = 1)$                                       |  |
|         | $P(E_1=1 \mathbf{T=0}, E_0=0)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=0)$ | $P(E_1=1 \mathbf{T=0}, E_0=1)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=1)$ |  |

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$                                                         | $E_0 = 1$                                                         |                    |
|---------|-------------------------------------------------------------------|-------------------------------------------------------------------|--------------------|
| $T = 0$ | $P(E_1 = 1 T = 0, E_0 = 0)$                                       | $P(E_1 = 1 T = 0, E_0 = 1)$                                       | $P(E_1 = 1 T = 0)$ |
| $T = 1$ | $P(E_1 = 1 T = 1, E_0 = 0)$                                       | $P(E_1 = 1 T = 1, E_0 = 1)$                                       | $P(E_1 = 1 T = 1)$ |
|         | $P(E_1=1 \mathbf{T=0}, E_0=0)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=0)$ | $P(E_1=1 \mathbf{T=0}, E_0=1)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=1)$ |                    |

# Predicción/estimación de impacto causal

con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$                                                         | $E_0 = 1$                                                         |                                                     |
|---------|-------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------|
| $T = 0$ | $P(E_1 = 1 T = 0, E_0 = 0)$                                       | $P(E_1 = 1 T = 0, E_0 = 1)$                                       | $P(E_1 = 1 T = 0)$                                  |
| $T = 1$ | $P(E_1 = 1 T = 1, E_0 = 0)$                                       | $P(E_1 = 1 T = 1, E_0 = 1)$                                       | $P(E_1 = 1 T = 1)$                                  |
|         | $P(E_1=1 \mathbf{T=0}, E_0=0)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=0)$ | $P(E_1=1 \mathbf{T=0}, E_0=1)$<br>$-P(E_1=1 \mathbf{T=1}, E_0=1)$ | $P(E_1=1 \mathbf{T=0})$<br>$-P(E_1=1 \mathbf{T=1})$ |

# Predicción/estimación de impacto causal

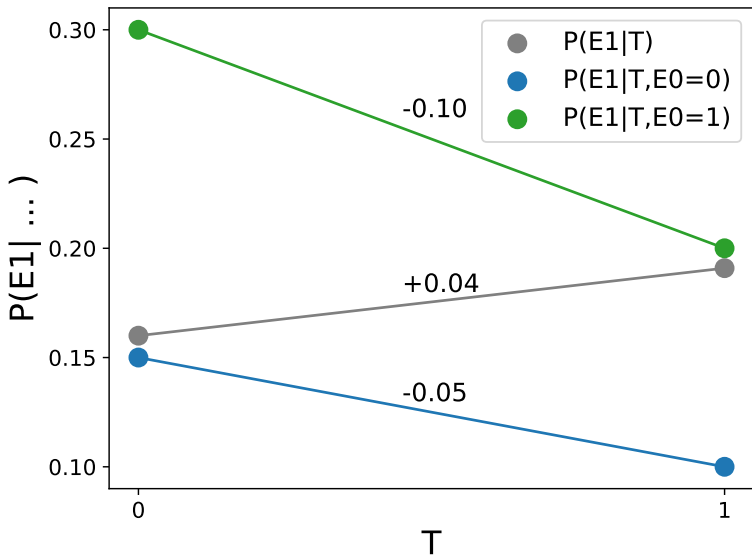
con datos observados sin intervenciones

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

|         | $E_0 = 0$       | $E_0 = 1$      |                 |
|---------|-----------------|----------------|-----------------|
| $T = 0$ | 15%<br>210/1400 | 30%<br>30/100  | 16%<br>240/1500 |
| $T = 1$ | 10%<br>5/50     | 20%<br>100/500 | 19%<br>105/550  |
|         | -5%             | -10%           | +4%             |

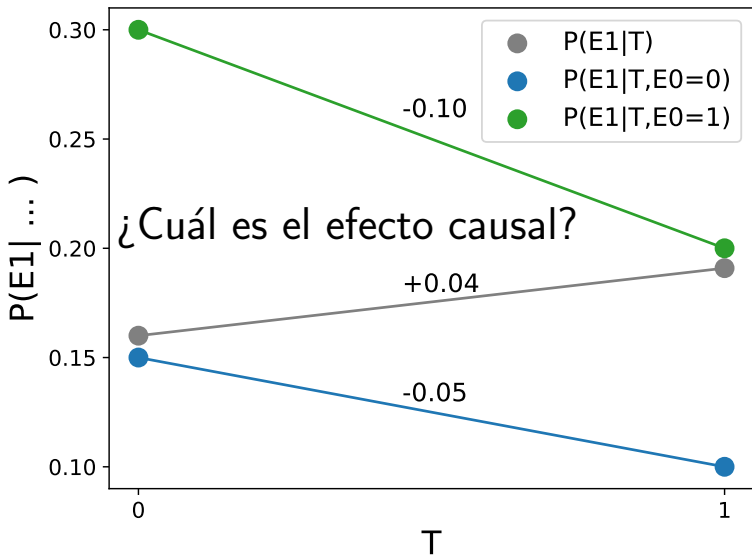
# Predicción/estimación de impacto causal

con datos observados sin intervenciones



# Predicción/estimación de impacto causal

con datos observados sin intervenciones



# Predicción/estimación de impacto causal

con datos observados sin intervenciones

Para predecir el impacto causal de las acciones  
**necesitamos conocer la estructura causal subyacente**

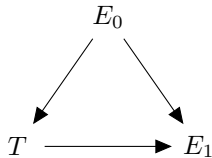
# Estimación de efecto causal

Modelo causal ( $M$ )

- Estado inicial:  $E_0 \in \{\text{Leve (0), Severo (1)}\}$
- Tratamiento:  $T \in \{\text{Básico (0), Especial (1)}\}$
- Estado final:  $E_1 \in \{\text{Leve (0), Severo (1)}\}$

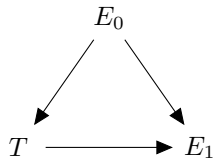
# Estimación de efecto causal

Modelo causal ( $M$ )



# Estimación de efecto causal

Modelo causal ( $M$ )



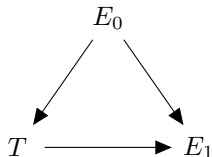
|         | $E_0 = 0$       | $E_0 = 1$      |                 |
|---------|-----------------|----------------|-----------------|
| $T = 0$ | 15%<br>210/1400 | 30%<br>30/100  | 16%<br>240/1500 |
| $T = 1$ | 10%<br>5/50     | 20%<br>100/500 | 19%<br>105/550  |

# Estimación de efecto causal

Modelo causal ( $M$ )

$P(E_0)$

| $E_0 = 0$ | $E_0 = 1$ |
|-----------|-----------|
| 1450/2050 | 600/2050  |



|         | $E_0 = 0$       | $E_0 = 1$      |                 |
|---------|-----------------|----------------|-----------------|
| $T = 0$ | 15%<br>210/1400 | 30%<br>30/100  | 16%<br>240/1500 |
| $T = 1$ | 10%<br>5/50     | 20%<br>100/500 | 19%<br>105/550  |

# Estimación de efecto causal

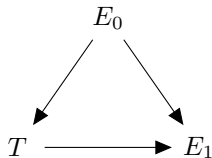
Modelo causal ( $M$ )

$P(E_0)$

| $E_0 = 0$ | $E_0 = 1$ |
|-----------|-----------|
| 1450/2050 | 600/2050  |

$P(T|E_0)$

|           | $T = 0$   | $T = 1$ |
|-----------|-----------|---------|
| $E_0 = 0$ | 1400/1450 | 50/1450 |
| $E_0 = 1$ | 100/600   | 500/600 |



|         | $E_0 = 0$       | $E_0 = 1$      |                 |
|---------|-----------------|----------------|-----------------|
| $T = 0$ | 15%<br>210/1400 | 30%<br>30/100  | 16%<br>240/1500 |
| $T = 1$ | 10%<br>5/50     | 20%<br>100/500 | 19%<br>105/550  |

# Estimación de efecto causal

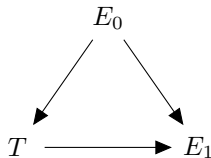
Modelo causal ( $M$ )

$P(E_0)$

| $E_0 = 0$ | $E_0 = 1$ |
|-----------|-----------|
| 1450/2050 | 600/2050  |

$P(T|E_0)$

|           | $T = 0$   | $T = 1$ |
|-----------|-----------|---------|
| $E_0 = 0$ | 1400/1450 | 50/1450 |
| $E_0 = 1$ | 100/600   | 500/600 |



$P(E_1|T, E_0)$

| $(E_0 = 0)$ | $E_1 = 0$ | $E_1 = 1$   |
|-------------|-----------|-------------|
| $T = 0$     | 0.85      | <b>0.15</b> |
| $T = 1$     | 0.90      | <b>0.10</b> |
| $(E_0 = 1)$ | $E_1 = 0$ | $E_1 = 1$   |
| $T = 0$     | 0.70      | <b>0.30</b> |
| $T = 1$     | 0.80      | <b>0.20</b> |

|         | $E_0 = 0$       | $E_0 = 1$      |                 |
|---------|-----------------|----------------|-----------------|
| $T = 0$ | 15%<br>210/1400 | 30%<br>30/100  | 16%<br>240/1500 |
| $T = 1$ | 10%<br>5/50     | 20%<br>100/500 | 19%<br>105/550  |

# Estimación de efecto causal

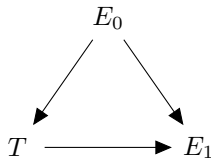
Modelo causal ( $M$ )

$P(E_0)$

| $E_0 = 0$ | $E_0 = 1$ |
|-----------|-----------|
| 1450/2050 | 600/2050  |

$P(T|E_0)$

|           | $T = 0$   | $T = 1$ |
|-----------|-----------|---------|
| $E_0 = 0$ | 1400/1450 | 50/1450 |
| $E_0 = 1$ | 100/600   | 500/600 |



$P(E_1|T, E_0)$

| $(E_0 = 0)$ | $E_1 = 0$ | $E_1 = 1$   |
|-------------|-----------|-------------|
| $T = 0$     | 0.85      | <b>0.15</b> |
| $T = 1$     | 0.90      | <b>0.10</b> |
| $(E_0 = 1)$ | $E_1 = 0$ | $E_1 = 1$   |
| $T = 0$     | 0.70      | <b>0.30</b> |
| $T = 1$     | 0.80      | <b>0.20</b> |

$P(E_1|T, E_0)$  es una predicción sin asociación espuria

un efecto causal *heterogéneo*, específico a cada estado inicial  $E_0$

# Estimación de efecto causal

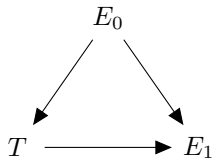
Modelo causal ( $M$ )

$P(E_0)$

| $E_0 = 0$ | $E_0 = 1$ |
|-----------|-----------|
| 1450/2050 | 600/2050  |

$P(T|E_0)$

|           | $T = 0$   | $T = 1$ |
|-----------|-----------|---------|
| $E_0 = 0$ | 1400/1450 | 50/1450 |
| $E_0 = 1$ | 100/600   | 500/600 |



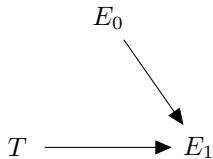
$P(E_1|T, E_0)$

| $(E_0 = 0)$ | $E_1 = 0$ | $E_1 = 1$   |
|-------------|-----------|-------------|
| $T = 0$     | 0.85      | <b>0.15</b> |
| $T = 1$     | 0.90      | <b>0.10</b> |
| $(E_0 = 1)$ | $E_1 = 0$ | $E_1 = 1$   |
| $T = 0$     | 0.70      | <b>0.30</b> |
| $T = 1$     | 0.80      | <b>0.20</b> |

$P(E_1|\text{do}(T))?$

# Estimación de efecto causal

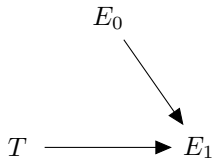
Modelo causal intervenido ( $M_T$ )



# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

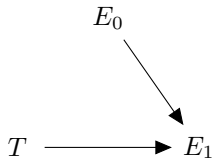


# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$



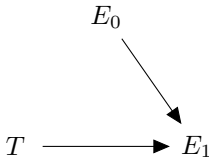
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



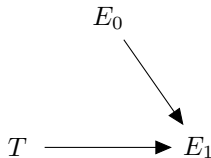
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



$$P(e_1|\text{do}(t)) = P_{M_T}(e_1|t)$$

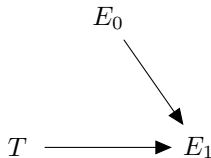
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



$$P(e_1|\text{do}(t)) = P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)}$$

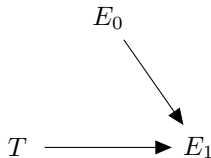
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



$$P(e_1|\text{do}(t)) = P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0, t, e_1)}{P_{M_T}(t)}$$

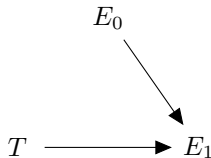
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



$$P(e_1|\text{do}(t)) = P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) P_{M_T}(t) P_{M_T}(e_1|t, e_0)}{P_{M_T}(t)}$$

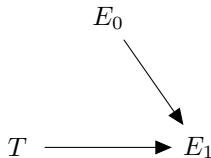
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



$$P(e_1|\text{do}(t)) = P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) \cancel{P_{M_T}(t)} P_{M_T}(e_1|t, e_0)}{\cancel{P_{M_T}(t)}}$$

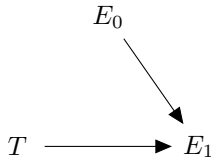
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



$$\begin{aligned} P(e_1|\text{do}(t)) &= P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) \cancel{P_{M_T}(t)} P_{M_T}(e_1|t, e_0)}{\cancel{P_{M_T}(t)}} \\ &= \sum_{e_0} P_{M_T}(e_0) P_{M_T}(e_1|t, e_0) \end{aligned}$$

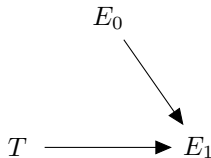
# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$P_{M_T}(T) = \text{Bern}(0.5)$$

$$P_{M_T}(E_0) = P(E_0)$$

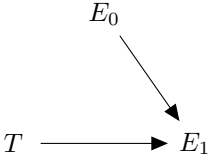
$$P_{M_T}(E_1|T, E_0) = P(E_1|T, E_0)$$



$$\begin{aligned} P(e_1|\text{do}(t)) &= P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) \cancel{P_{M_T}(t)} P_{M_T}(e_1|t, e_0)}{\cancel{P_{M_T}(t)}} \\ &= \sum_{e_0} P(e_0) P(e_1|t, e_0) \end{aligned}$$

# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

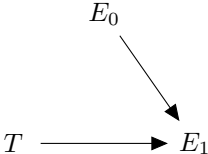
$$\begin{aligned} P_{M_T}(T) &= \text{Bern}(0.5) \\ P_{M_T}(E_0) &= P(E_0) \\ P_{M_T}(E_1|T, E_0) &= P(E_1|T, E_0) \end{aligned}$$


```
graph LR; E0 --> E1; T --> E1;
```

$$\begin{aligned} P(e_1|\text{do}(t)) &= P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) \cancel{P_{M_T}(t)} P_{M_T}(e_1|t, e_0)}{\cancel{P_{M_T}(t)}} \\ &= \underbrace{\sum_{e_0} P(e_0) P(e_1|t, e_0)}_{\text{Distribuciones observadas!}} \end{aligned}$$

# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$\begin{aligned} P_{M_T}(T) &= \text{Bern}(0.5) \\ P_{M_T}(E_0) &= P(E_0) \\ P_{M_T}(E_1|T, E_0) &= P(E_1|T, E_0) \end{aligned}$$


```
graph LR; E0 --> E1; T --> E1;
```

$$P(e_1|\text{do}(t)) = P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) \cancel{P_{M_T}(t)} P_{M_T}(e_1|t, e_0)}{\cancel{P_{M_T}(t)}}$$

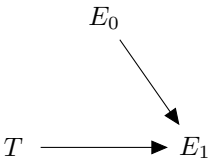
$$= \underbrace{\sum_{e_0} P(e_0) P(e_1|t, e_0)}$$

Distribuciones  
observadas!

- Encontramos los efectos causales en cada subgrupo  $P(e_1|t, e_0)$
- Y los ponderamos por el tamaño de cada subgrupo  $P(e_0)$

# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

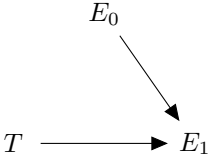
$$\begin{aligned} P_{M_T}(T) &= \text{Bern}(0.5) \\ P_{M_T}(E_0) &= P(E_0) \\ P_{M_T}(E_1|T, E_0) &= P(E_1|T, E_0) \end{aligned}$$


$$\begin{aligned} P(e_1|\text{do}(t)) &= P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) \cancel{P_{M_T}(t)} P_{M_T}(e_1|t, e_0)}{\cancel{P_{M_T}(t)}} \\ &= \sum_{e_0} P(e_0) P(e_1|t, e_0) \end{aligned}$$

$$\underbrace{P(E_1 = 1|\text{do}(T = 1)) - P(E_1 = 1|\text{do}(T = 0))}_{\text{Efecto causal general}}$$

# Estimación de efecto causal

Modelo causal intervenido ( $M_T$ )

$$\begin{aligned} P_{M_T}(T) &= \text{Bern}(0.5) \\ P_{M_T}(E_0) &= P(E_0) \\ P_{M_T}(E_1|T, E_0) &= P(E_1|T, E_0) \end{aligned}$$


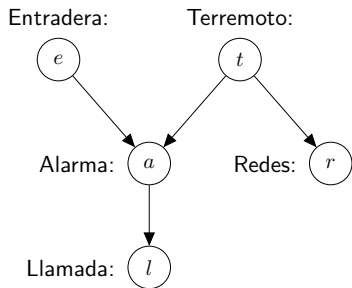
```
graph LR; E0 --> E1; T --> E1;
```

$$\begin{aligned} P(e_1|\text{do}(t)) &= P_{M_T}(e_1|t) = \frac{P_{M_T}(e_1, t)}{P_{M_T}(t)} = \frac{\sum_{e_0} P_{M_T}(e_0) \cancel{P_{M_T}(t)} P_{M_T}(e_1|t, e_0)}{\cancel{P_{M_T}(t)}} \\ &= \sum_{e_0} P(e_0) P(e_1|t, e_0) \end{aligned}$$

$$\underbrace{P(E_1 = 1|\text{do}(T = 1)) - P(E_1 = 1|\text{do}(T = 0))}_{\text{Efecto causal general}} = -0.0646$$

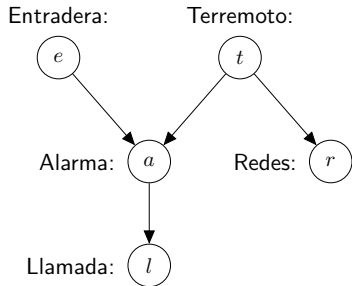
# Flujo de inferencia

## Estructuras básicas



# Flujo de inferencia

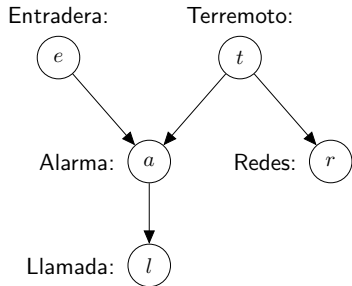
## Estructuras básicas



Pipe:  $e \rightarrow a \rightarrow l$

# Flujo de inferencia

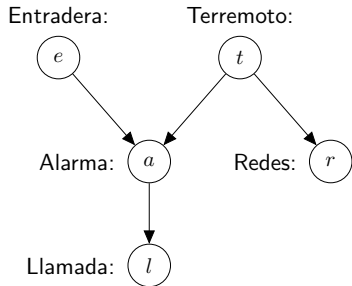
## Estructuras básicas



Pipe:  $e \rightarrow a \rightarrow l$        $P(l) \stackrel{?}{=} P(l|e)$

# Flujo de inferencia

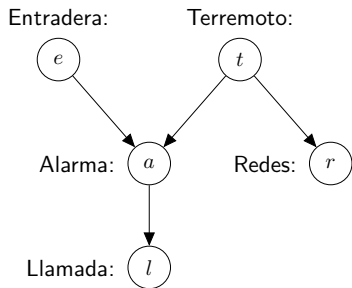
## Estructuras básicas



Pipe:  $e \rightarrow a \rightarrow l$        $P(l) \stackrel{?}{=} P(l|e)$        $P(l|a) \stackrel{?}{=} P(l|e, a)$

# Flujo de inferencia

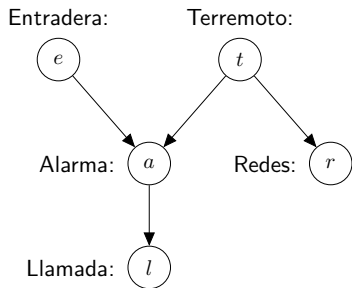
## Estructuras básicas



|       | Intermedio<br>no observable     | Intermedio<br>observable      |                                    |
|-------|---------------------------------|-------------------------------|------------------------------------|
| Pipe: | $e \rightarrow a \rightarrow l$ | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$ |

# Flujo de inferencia

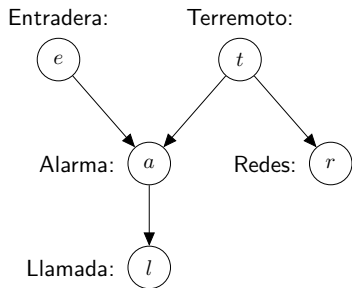
## Estructuras básicas



|       |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|-------|---------------------------------|-------------------------------|------------------------------------|
| Pipe: | $e \rightarrow a \rightarrow l$ | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$ |
| Fork: | $a \leftarrow t \rightarrow r$  |                               |                                    |

# Flujo de inferencia

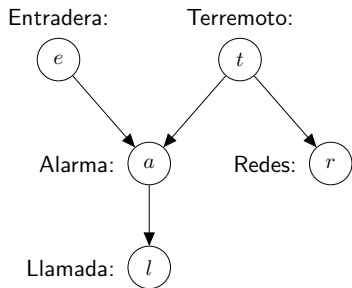
## Estructuras básicas



|       |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|-------|---------------------------------|-------------------------------|------------------------------------|
| Pipe: | $e \rightarrow a \rightarrow l$ | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$ |
| Fork: | $a \leftarrow t \rightarrow r$  | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$ |

# Flujo de inferencia

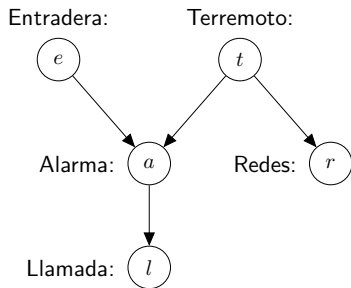
## Estructuras básicas



|           |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|-----------|---------------------------------|-------------------------------|------------------------------------|
| Pipe:     | $e \rightarrow a \rightarrow l$ | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$ |
| Fork:     | $a \leftarrow t \rightarrow r$  | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$ |
| Collider: | $e \rightarrow a \leftarrow t$  |                               |                                    |

# Flujo de inferencia

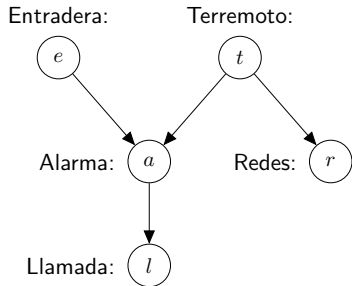
## Estructuras básicas



|           |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|-----------|---------------------------------|-------------------------------|------------------------------------|
| Pipe:     | $e \rightarrow a \rightarrow l$ | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$ |
| Fork:     | $a \leftarrow t \rightarrow r$  | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$ |
| Collider: | $e \rightarrow a \leftarrow t$  | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$ |

# Flujo de inferencia

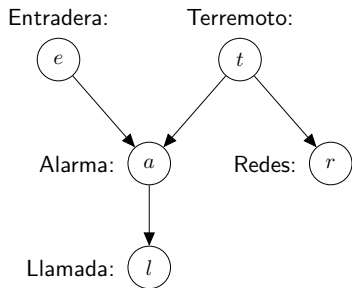
## Estructuras básicas



|           |                                                       | Intermedio<br>no observable   | Intermedio<br>observable                                                 |
|-----------|-------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------|
| Pipe:     | $e \rightarrow a \rightarrow l$                       | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$                                       |
| Fork:     | $a \leftarrow t \rightarrow r$                        | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$                                       |
| Collider: | $e \rightarrow a \leftarrow t$<br>$\downarrow$<br>$l$ | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$<br>$P(t l) \stackrel{?}{=} P(t e, l)$ |

# Flujo de inferencia

## Estructuras básicas



|                  |                                                       | Intermedio<br>no observable   | Intermedio<br>observable                                                 |
|------------------|-------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------|
| <b>Pipe:</b>     | $e \rightarrow a \rightarrow l$                       | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$                                       |
| <b>Fork:</b>     | $a \leftarrow t \rightarrow r$                        | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$                                       |
| <b>Collider:</b> | $e \rightarrow a \leftarrow t$<br>$\downarrow$<br>$l$ | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$<br>$P(t l) \stackrel{?}{=} P(t e, l)$ |

# Flujo de inferencia

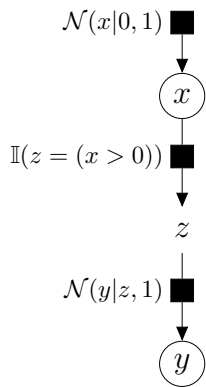
Pipe



# Flujo de inferencia

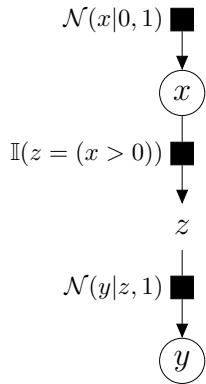
## Pipe

$$x \longrightarrow z \longrightarrow y$$

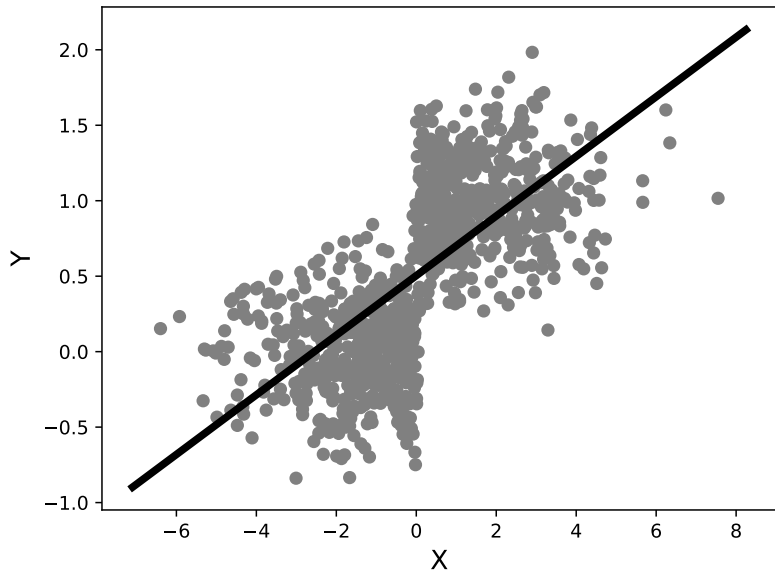


# Flujo de inferencia

## Pipe

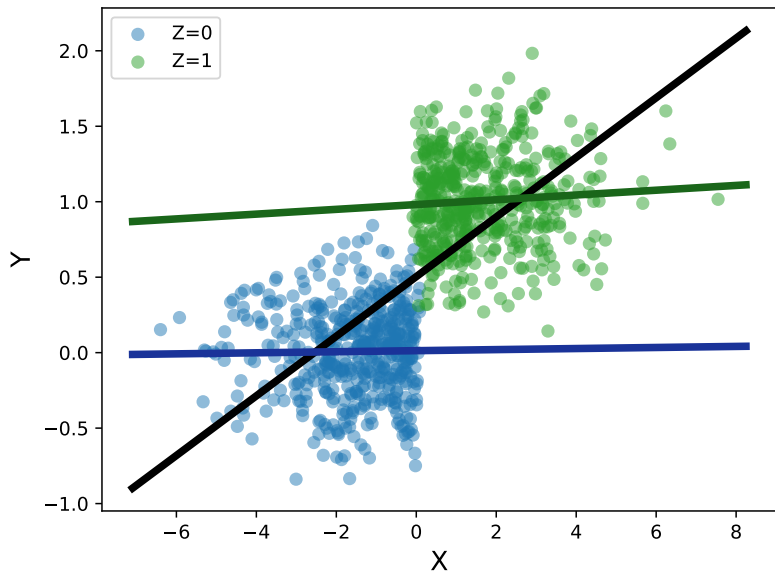
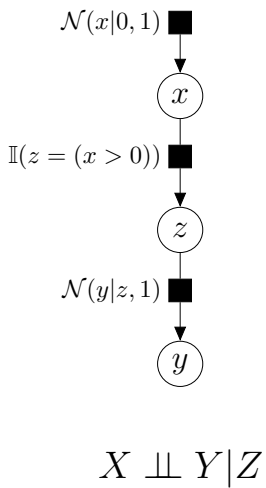


$$X \not\perp Y | \emptyset$$



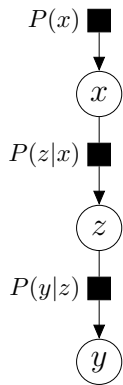
# Flujo de inferencia

## Pipe



# Flujo de inferencia

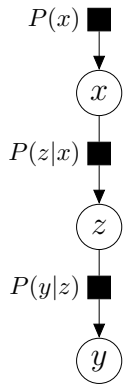
## Pipe



$$X \perp\!\!\!\perp Y|Z \iff P(x, y|z) = P(x|z)P(y|z)$$

# Flujo de inferencia

## Pipe

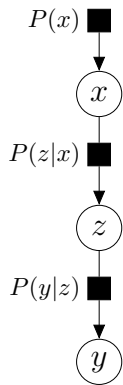


$$P(x, y|z) =$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

## Pipe

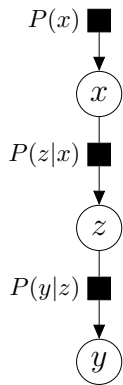


$$P(x, y|z) = \frac{P(x, y, z)}{P(z)}$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

## Pipe

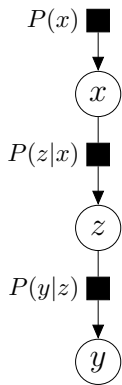


$$P(x, y|z) = \frac{P(x)P(z|x)P(y|z)}{P(z)}$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

## Pipe

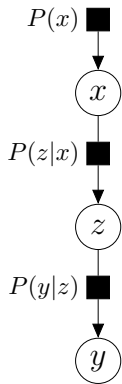


$$P(x, y|z) = \frac{P(x)P(z|x)P(y|z)}{P(z)} = \frac{P(x)P(z|x)}{P(z)}P(y|z)$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

## Pipe

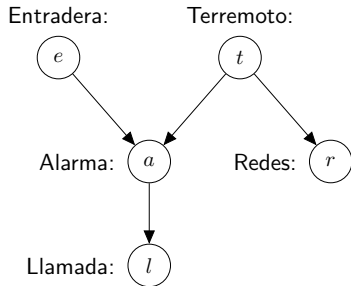


$$P(x, y|z) = \frac{P(x)P(z|x)P(y|z)}{P(z)} = \frac{P(x)P(z|x)}{P(z)}P(y|z) = P(x|z)P(y|z)$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

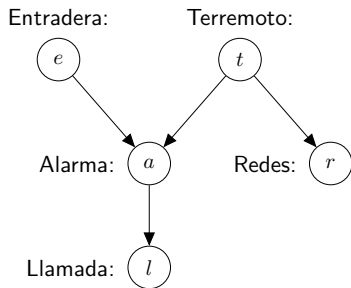
## Estructuras básicas



|                  |                                                       | Intermedio<br>no observable   | Intermedio<br>observable                                                     |
|------------------|-------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------|
| <b>Pipe :</b>    | $e \rightarrow a \rightarrow l$                       | $P(l) \stackrel{?}{=} P(l e)$ | $P(l a) \stackrel{?}{=} P(l e, a)$                                           |
| <b>Fork:</b>     | $a \leftarrow t \rightarrow r$                        | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$                                           |
| <b>Collider:</b> | $e \rightarrow a \leftarrow t$<br>$\downarrow$<br>$l$ | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$<br><br>$P(t l) \stackrel{?}{=} P(t e, l)$ |

# Flujo de inferencia

## Estructuras básicas

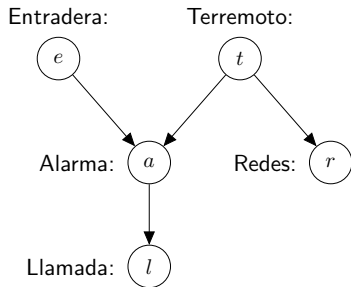


|                  |                                                       | Intermedio<br>no observable   | Intermedio<br>observable                                                     |
|------------------|-------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------|
| <b>Pipe :</b>    | $e \rightarrow a \rightarrow l$                       | $P(l) \neq P(l e)$            | $P(l a) \stackrel{?}{=} P(l e, a)$                                           |
| <b>Fork:</b>     | $a \leftarrow t \rightarrow r$                        | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$                                           |
| <b>Collider:</b> | $e \rightarrow a \leftarrow t$<br>$\downarrow$<br>$l$ | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$<br><br>$P(t l) \stackrel{?}{=} P(t e, l)$ |

$$E \not\perp L | \emptyset$$

# Flujo de inferencia

## Estructuras básicas



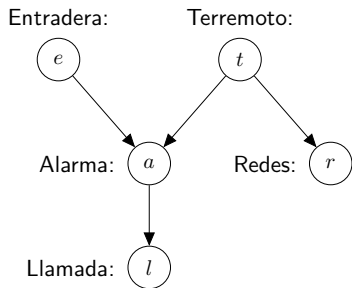
|                  |                                                       | Intermedio<br>no observable   | Intermedio<br>observable                                                 |
|------------------|-------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------|
| <b>Pipe :</b>    | $e \rightarrow a \rightarrow l$                       | $P(l) \neq P(l e)$            | $P(l a) = P(l e, a)$                                                     |
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| <b>Collider:</b> | $e \rightarrow a \leftarrow t$<br>$\downarrow$<br>$l$ | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$<br>$P(t l) \stackrel{?}{=} P(t e, l)$ |

$$E \not\perp\!\!\!\perp L | \emptyset$$

$$E \perp\!\!\!\perp L | A$$

# Flujo de inferencia

## Estructuras básicas



|              |                                                       | Intermedio<br>no observable   | Intermedio<br>observable                                                     |
|--------------|-------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------|
| Pipe :       | $e \rightarrow a \rightarrow l$                       | $P(l) \neq P(l e)$            | $P(l a) = P(l e, a)$                                                         |
| <b>Fork:</b> | $a \leftarrow t \rightarrow r$                        | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$                                           |
| Collider:    | $e \rightarrow a \leftarrow t$<br>$\downarrow$<br>$l$ | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$<br><br>$P(t l) \stackrel{?}{=} P(t e, l)$ |

# Flujo de inferencia

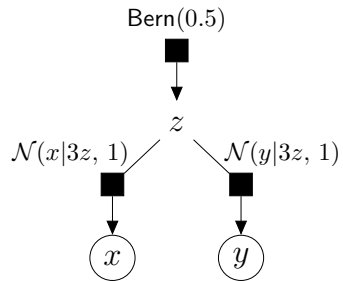
Fork

$$x \longleftarrow z \longrightarrow y$$

# Flujo de inferencia

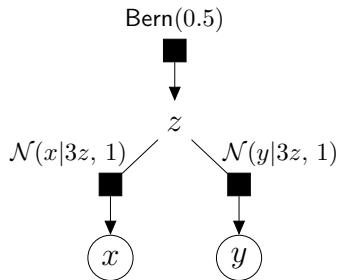
Fork

$$x \longleftarrow z \longrightarrow y$$

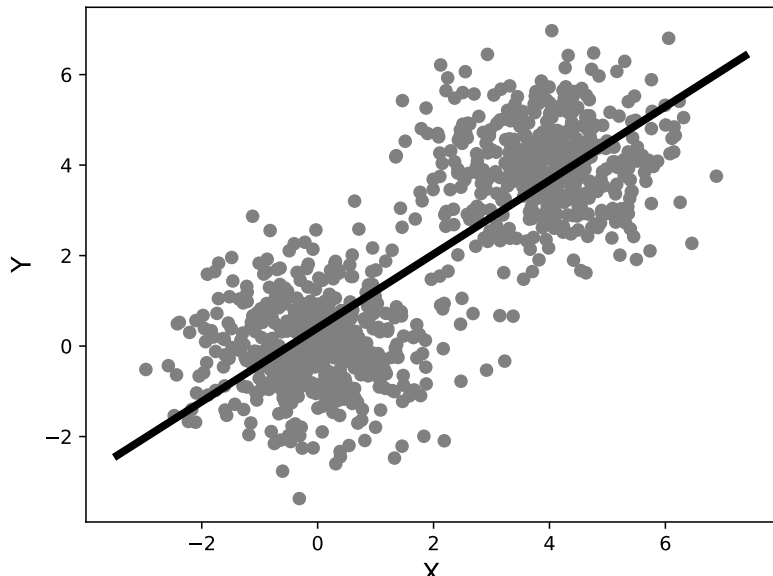


# Flujo de inferencia

## Fork

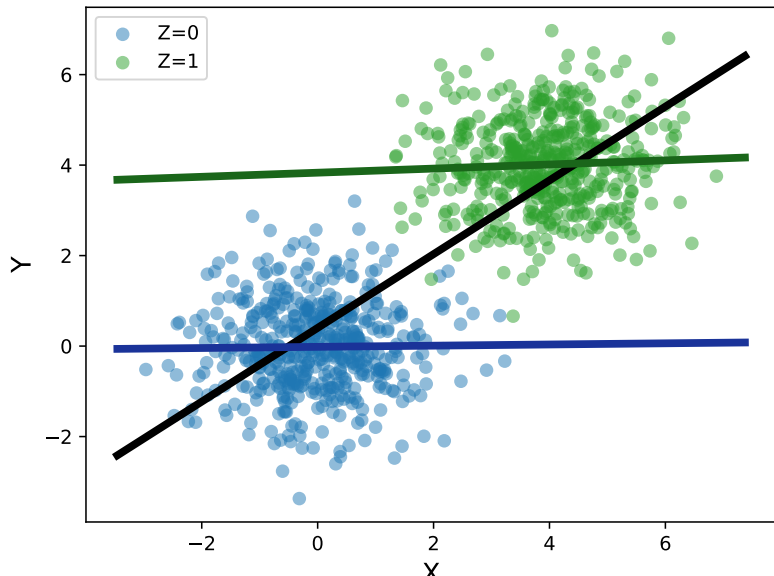
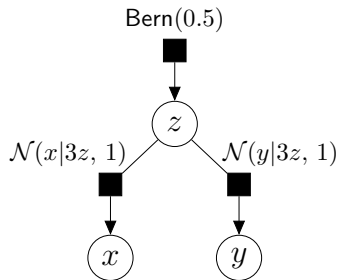


$$X \not\perp Y | \emptyset$$



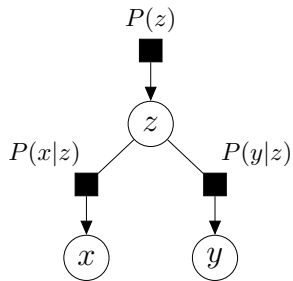
# Flujo de inferencia

## Fork



# Flujo de inferencia

## Fork

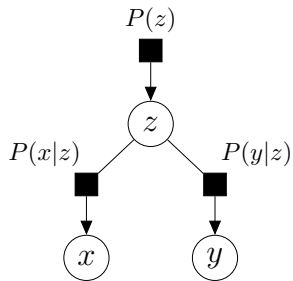


$$P(x, y|z) =$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

## Fork

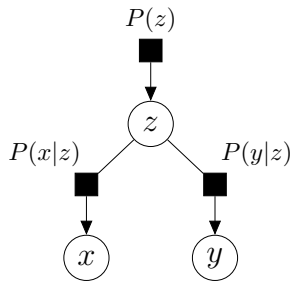


$$P(x, y|z) = \frac{P(x, y, z)}{P(z)}$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

## Fork

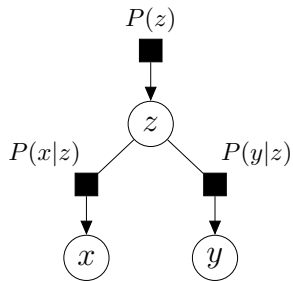


$$P(x, y|z) = \frac{P(x|z)P(y|z)P(z)}{P(z)}$$

$$X \perp\!\!\!\perp Y|Z$$

# Flujo de inferencia

## Fork

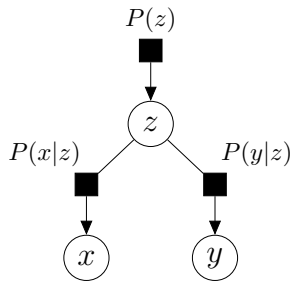


$$X \perp\!\!\!\perp Y | Z$$

$$P(x, y|z) = \frac{P(x|z)P(y|z)\cancel{P(z)}}{\cancel{P(z)}}$$

# Flujo de inferencia

## Fork

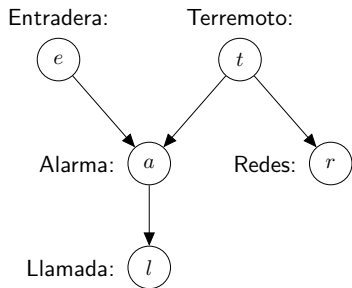


$$X \perp\!\!\!\perp Y|Z$$

$$P(x, y|z) = \frac{P(x|z)P(y|z)\cancel{P(z)}}{\cancel{P(z)}} = P(x|z)P(y|z)$$

# Flujo de inferencia

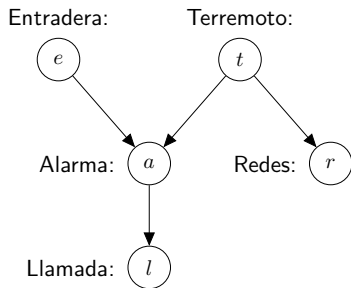
## Estructuras básicas



|              |                                                       | Intermedio<br>no observable   | Intermedio<br>observable                                                     |
|--------------|-------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------|
| Pipe:        | $e \rightarrow a \rightarrow l$                       | $P(l) \neq P(l e)$            | $P(l a) = P(l e, a)$                                                         |
| <b>Fork:</b> | $a \leftarrow t \rightarrow r$                        | $P(r) \stackrel{?}{=} P(r a)$ | $P(r t) \stackrel{?}{=} P(r t, a)$                                           |
| Collider:    | $e \rightarrow a \leftarrow t$<br>$\downarrow$<br>$l$ | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$<br><br>$P(t l) \stackrel{?}{=} P(t e, l)$ |

# Flujo de inferencia

## Estructuras básicas

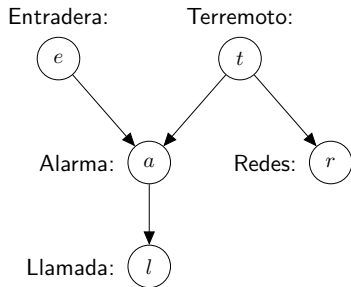


|              |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|--------------|---------------------------------|-------------------------------|------------------------------------|
| Pipe:        | $e \rightarrow a \rightarrow l$ | $P(l) \neq P(l e)$            | $P(l a) = P(l e, a)$               |
| <b>Fork:</b> | $a \leftarrow t \rightarrow r$  | $P(r) \neq P(r a)$            | $P(r t) \stackrel{?}{=} P(r t, a)$ |
| Collider:    | $e \rightarrow a \leftarrow t$  | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$ |
|              | $\downarrow$<br>$l$             |                               | $P(t l) \stackrel{?}{=} P(t e, l)$ |

$$A \not\perp R | \emptyset$$

# Flujo de inferencia

## Estructuras básicas



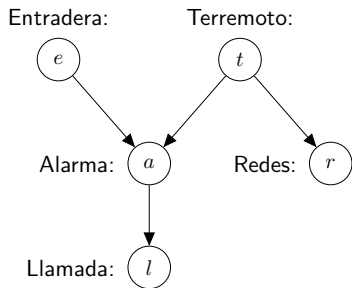
|              |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|--------------|---------------------------------|-------------------------------|------------------------------------|
| Pipe:        | $e \rightarrow a \rightarrow l$ | $P(l) \neq P(l e)$            | $P(l a) = P(l e, a)$               |
| <b>Fork:</b> | $a \leftarrow t \rightarrow r$  | $P(r) \neq P(r a)$            | $P(r t) = P(r t, a)$               |
| Collider:    | $e \rightarrow a \leftarrow t$  | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$ |
|              | $\downarrow$<br>$l$             |                               | $P(t l) \stackrel{?}{=} P(t e, l)$ |

$$A \not\perp\!\!\!\perp R|\emptyset$$

$$A \perp\!\!\!\perp R|T$$

# Flujo de inferencia

## Estructuras básicas



|           |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|-----------|---------------------------------|-------------------------------|------------------------------------|
| Pipe:     | $e \rightarrow a \rightarrow l$ | $P(l) \neq P(l e)$            | $P(l a) = P(l e, a)$               |
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| Collider: | $e \rightarrow a \leftarrow t$  | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$ |
|           | $\downarrow$<br>$l$             |                               | $P(t l) \stackrel{?}{=} P(t e, l)$ |

# Flujo de inferencia

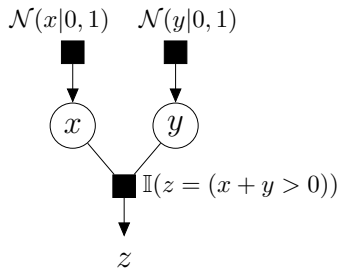
## Collider



# Flujo de inferencia

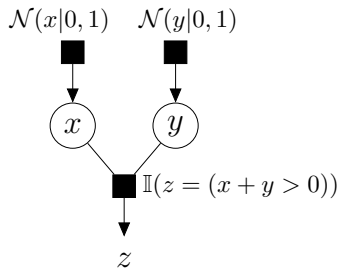
## Collider

$$x \longrightarrow z \longleftarrow y$$

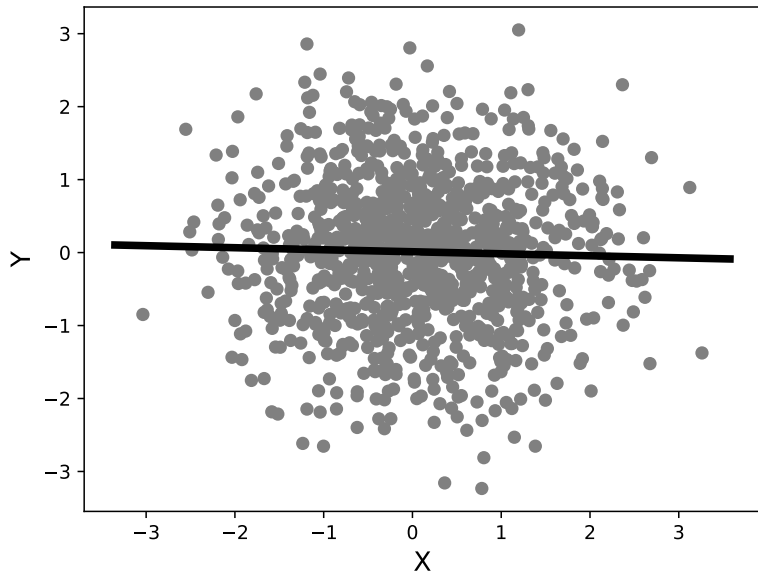


# Flujo de inferencia

## Collider

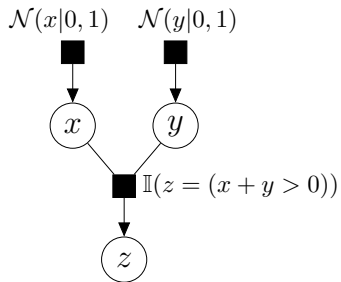


$$X \perp\!\!\!\perp Y | \emptyset$$

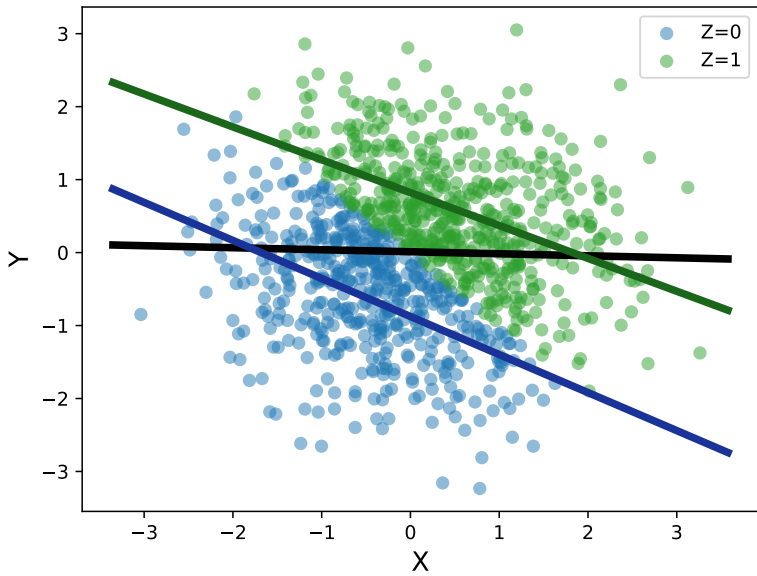


# Flujo de inferencia

## Collider

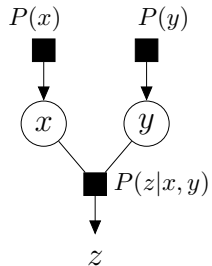


$$X \not\perp\!\!\!\perp Y|Z$$



# Flujo de inferencia

## Collider

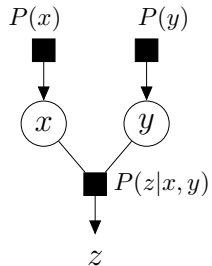


$$P(x, y) =$$

$$X \perp\!\!\!\perp Y | \emptyset$$

# Flujo de inferencia

## Collider

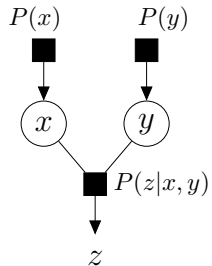


$$P(x, y) = \sum_z P(x, y, z)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

# Flujo de inferencia

## Collider

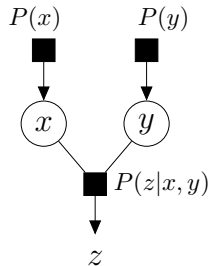


$$P(x, y) = \sum_z P(x)P(y)P(z|x, y)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

# Flujo de inferencia

## Collider

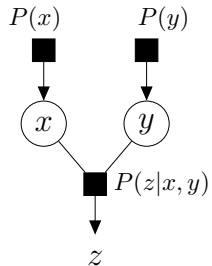


$$P(x, y) = \sum_z P(x)P(y)P(z|x, y) = P(x)P(y) \sum_z P(z|x, y)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

# Flujo de inferencia

## Collider

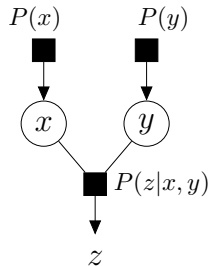


$$P(x, y) = \sum_z P(x)P(y)P(z|x, y) = P(x)P(y) \cancel{\sum_z P(z|x, y)}$$

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# Flujo de inferencia

## Collider

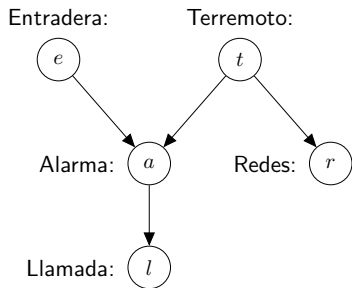


$$P(x, y) = \sum_z P(x)P(y)P(z|x, y) = P(x)P(y)$$

$$X \perp\!\!\!\perp Y | \emptyset$$

# Flujo de inferencia

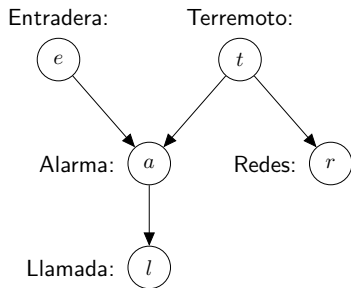
## Estructuras básicas



|           |                                 | Intermedio<br>no observable   | Intermedio<br>observable           |
|-----------|---------------------------------|-------------------------------|------------------------------------|
| Pipe:     | $e \rightarrow a \rightarrow l$ | $P(l) \neq P(l e)$            | $P(l a) = P(l e, a)$               |
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| Collider: | $e \rightarrow a \leftarrow t$  | $P(t) \stackrel{?}{=} P(t e)$ | $P(t a) \stackrel{?}{=} P(t e, a)$ |
|           | $\downarrow$<br>$\bar{l}$       |                               | $P(t l) \stackrel{?}{=} P(t e, l)$ |

# Flujo de inferencia

## Estructuras básicas

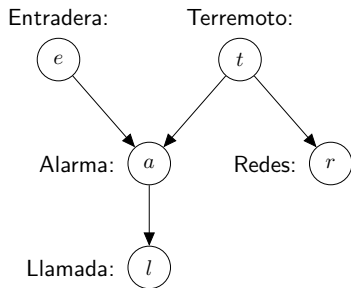


|           |                                 | Intermedio<br>no observable | Intermedio<br>observable           |
|-----------|---------------------------------|-----------------------------|------------------------------------|
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| Collider: | $e \rightarrow a \leftarrow t$  | $P(t) = P(t e)$             | $P(t a) \stackrel{?}{=} P(t e, a)$ |
|           | $\downarrow$<br>$\bar{l}$       |                             | $P(t l) \stackrel{?}{=} P(t e, l)$ |

$$T \perp\!\!\!\perp E | \emptyset$$

# Flujo de inferencia

## Estructuras básicas



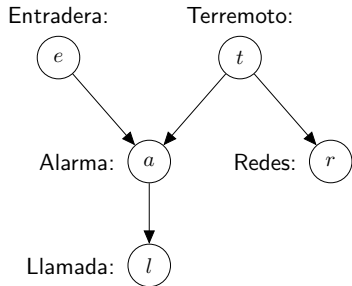
|           |                                 | Intermedio<br>no observable | Intermedio<br>observable           |
|-----------|---------------------------------|-----------------------------|------------------------------------|
| Pipe:     | $e \rightarrow a \rightarrow l$ | $P(l) \neq P(l e)$          | $P(l a) = P(l e, a)$               |
| Fork:     | $a \leftarrow t \rightarrow r$  | $P(r) \neq P(r a)$          | $P(r t) = P(r t, a)$               |
| Collider: | $e \rightarrow a \leftarrow t$  | $P(t) = P(t e)$             | $P(t a) \neq P(t e, a)$            |
|           | $\downarrow$<br>$\bar{l}$       |                             | $P(t l) \stackrel{?}{=} P(t e, l)$ |

$$T \perp\!\!\!\perp E | \emptyset$$

$$T \not\perp\!\!\!\perp E | A$$

# Flujo de inferencia

## Estructuras básicas



|           |                                 | Intermedio<br>no observable | Intermedio<br>observable |
|-----------|---------------------------------|-----------------------------|--------------------------|
| Pipe:     | $e \rightarrow a \rightarrow l$ | $P(l) \neq P(l e)$          | $P(l a) = P(l e, a)$     |
| Fork:     | $a \leftarrow t \rightarrow r$  | $P(r) \neq P(r a)$          | $P(r t) = P(r t, a)$     |
| Collider: | $e \rightarrow a \leftarrow t$  | $P(t) = P(t e)$             | $P(t a) \neq P(t e, a)$  |
|           | $\downarrow$<br>$\bar{l}$       |                             | $P(t l) \neq P(t e, l)$  |

$$T \perp\!\!\!\perp E | \emptyset$$

$$T \not\perp\!\!\!\perp E | L$$

# Flujo de inferencia

## Estructuras complejas (caminos)

Hay flujo de inferencia entre los extremos de una cadena si:  
(camino *d-conectado*)

- Todas las consecuencias comunes (o sus descendientes) son observables
- Ninguna otra variable es observable

# Flujo de inferencia

## Estructuras complejas (caminos)

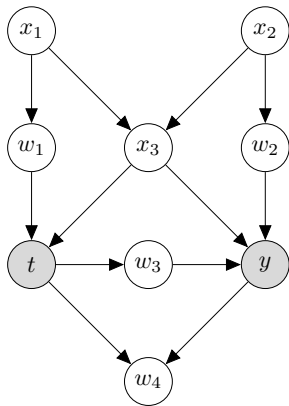
Hay flujo de inferencia entre los extremos de una cadena si:  
(camino *d-conectado*)

- Todas las consecuencias comunes (o sus descendientes) son observables
- Ninguna otra variable es observable

Se cierra el flujo si está no *d-conectado*  
d-separado

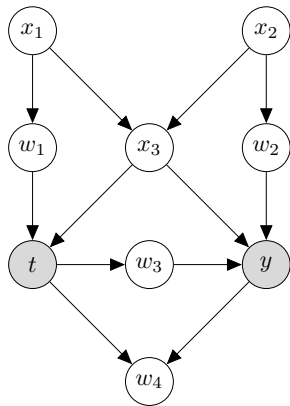
# Flujo de inferencia

## Estructuras complejas (caminos)



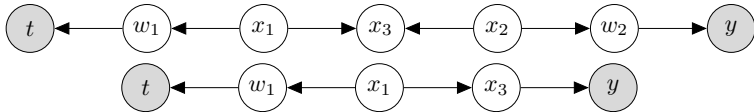
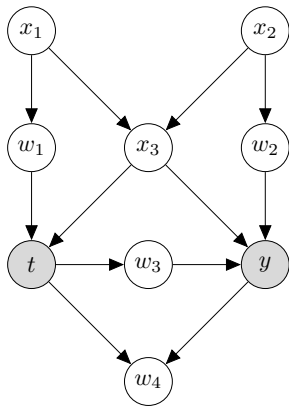
# Flujo de inferencia

## Estructuras complejas (caminos)



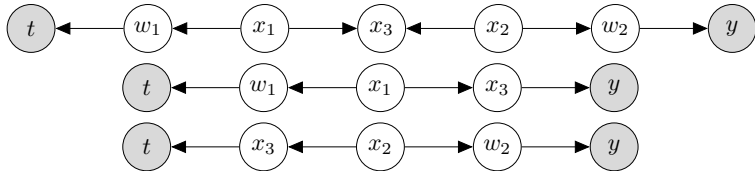
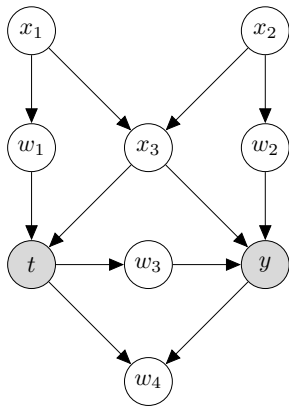
# Flujo de inferencia

## Estructuras complejas (caminos)



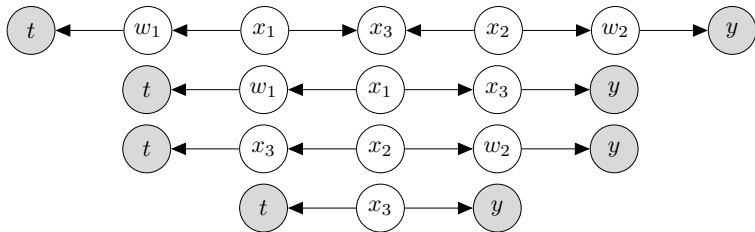
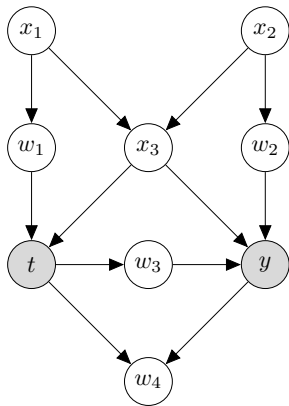
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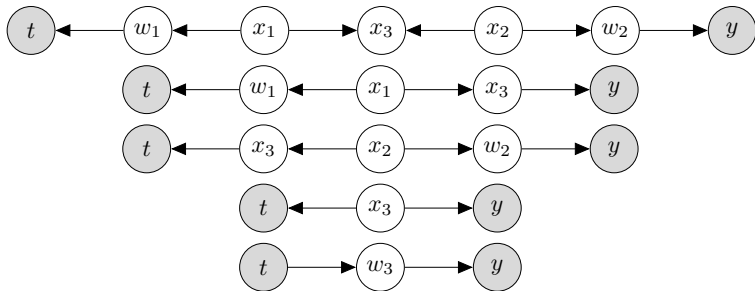
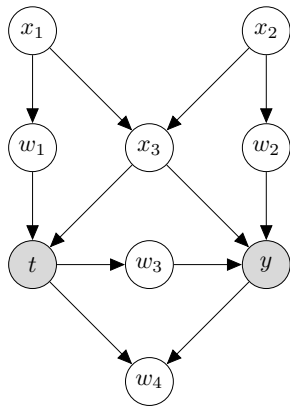
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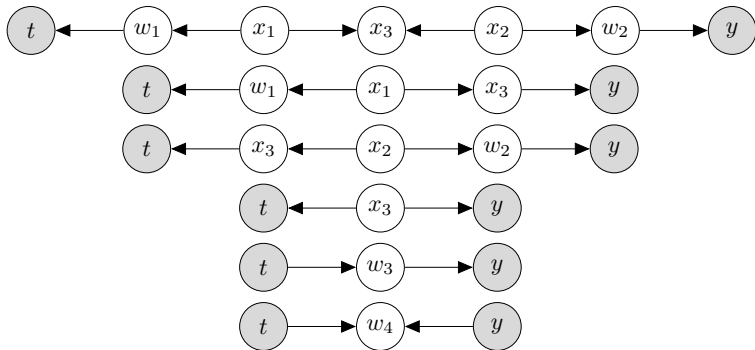
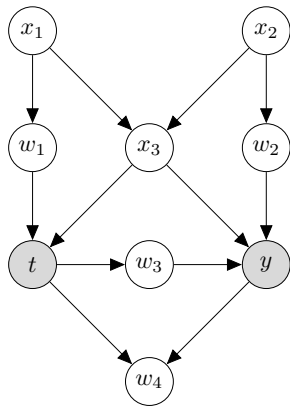
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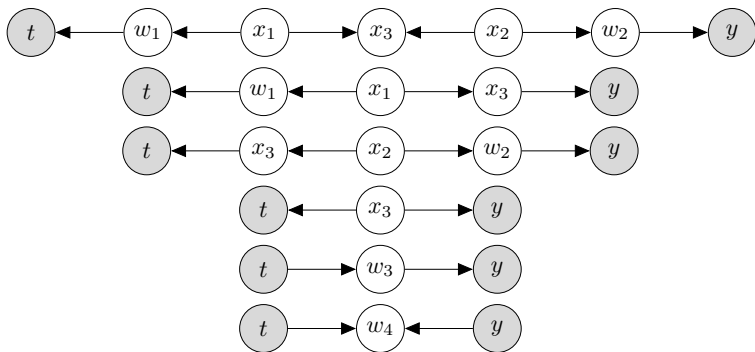
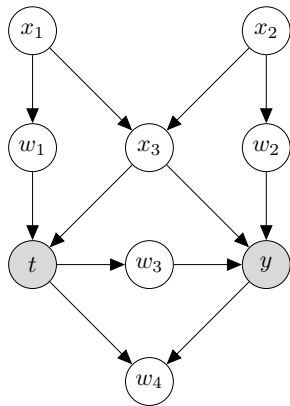
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# Flujo de inferencia

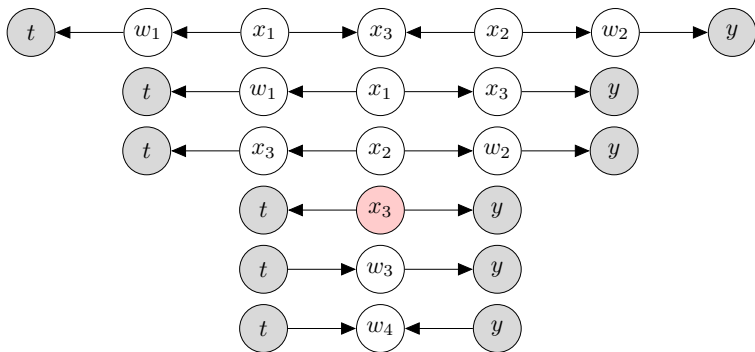
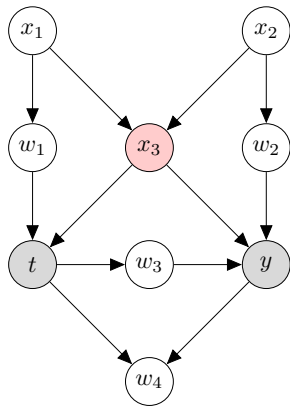
## Estructuras complejas (caminos)



Para hacer predicciones “causales” necesitamos eliminar el flujo de inferencia que circula por los caminos no causales

# Flujo de inferencia

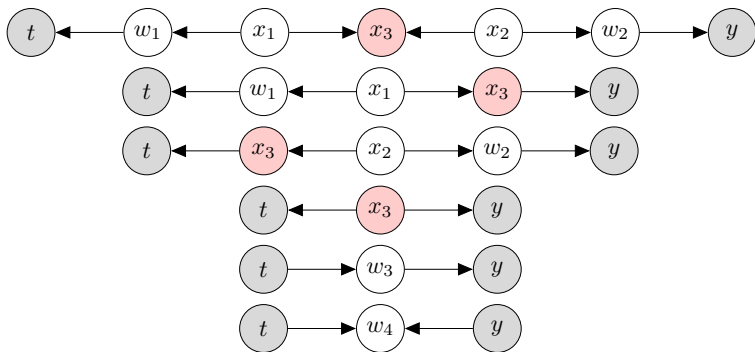
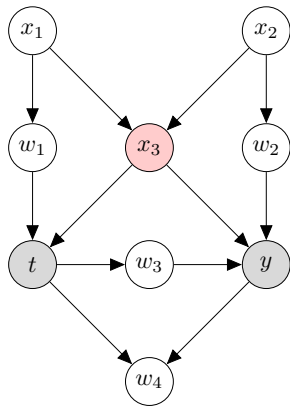
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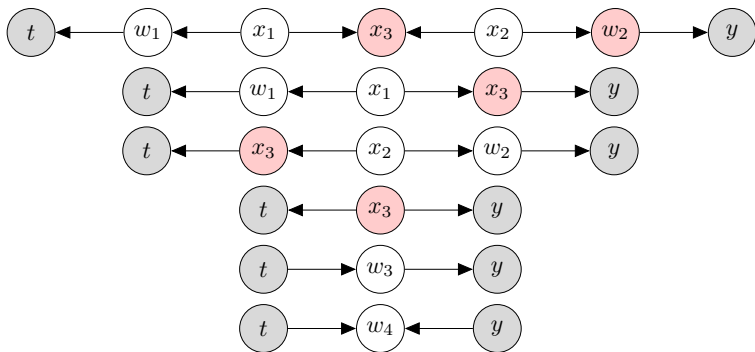
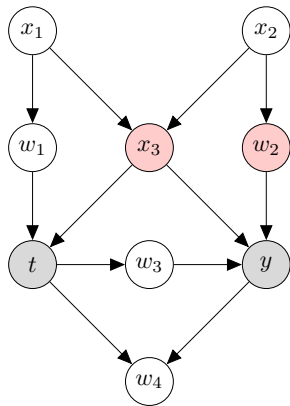
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Para hacer predicciones “causales” necesitamos eliminar el flujo de inferencia que circula por los caminos no causales

# Flujo de inferencia

## Estructuras complejas (caminos)



Para hacer predicciones “causales” necesitamos eliminar el flujo de inferencia que circula por los caminos no causales

# Flujo de inferencia

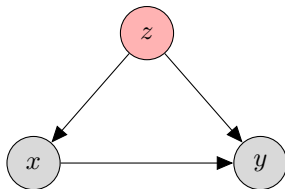
## Variables de control

### Backdoor criterion

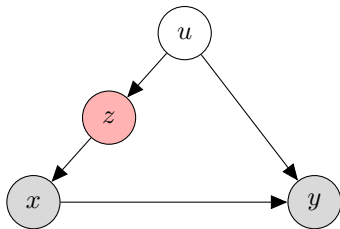
Dado un conjunto de variable  $Q$  tal que:

1.  $Q$  cierra todos los caminos traseros de  $T$  a  $Y$
2.  $Q$  no contiene ningún descendiente de  $T$

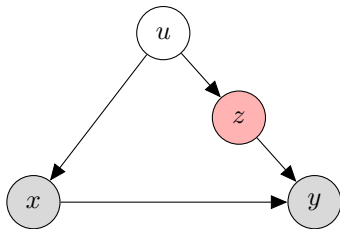
# Controles buenos



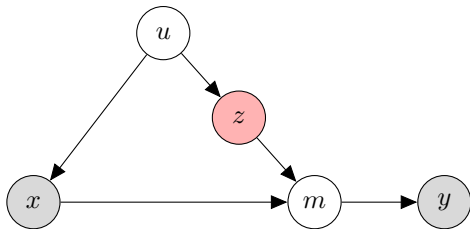
# Controles buenos



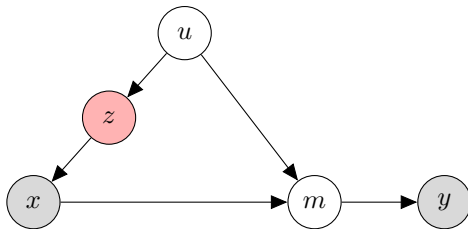
# Controles buenos



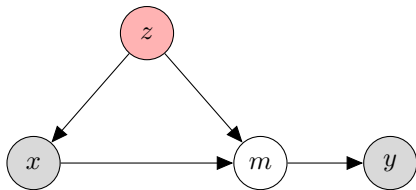
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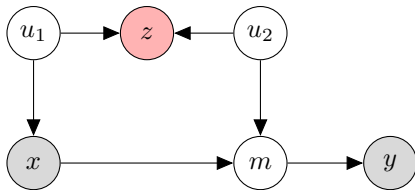
# Controles buenos



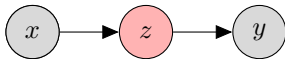
# Controles buenos



# Controles malos

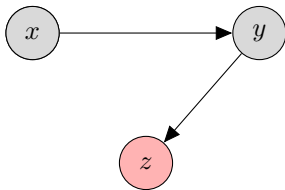


# Controles malos



# Controles malos

## Sesgo de selección

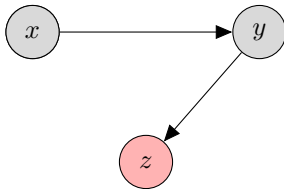


# Controles malos

## Sesgo de selección

Caso borde  $Y = Z$ .

$$\mathbb{E}[Y|T, Z] = \mathbb{E}[Y|T, Y]$$

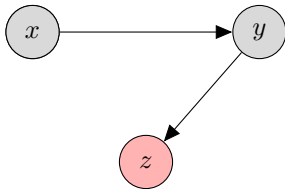


# Controles malos

## Sesgo de selección

Caso borde  $Y = Z$ .  $T$  se hace independiente de  $Y$ .

$$\mathbb{E}[Y|T, Z] = \mathbb{E}[Y|T, Y] = \mathbb{E}[Y|Y] = Y$$

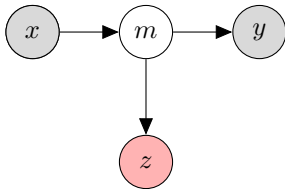


# Controles malos

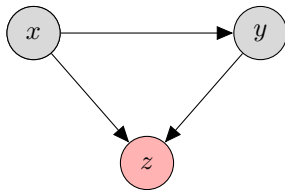
## Sesgo de selección

Caso borde  $M = Z$ .  $T$  se hace independiente de  $Y$ .

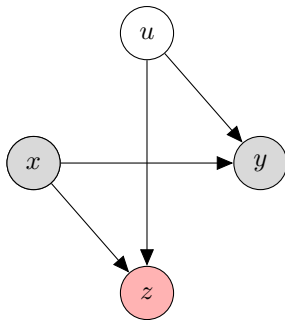
$$\mathbb{E}[Y|T, Z] = \mathbb{E}[Y|T, M]$$



# Controles malos

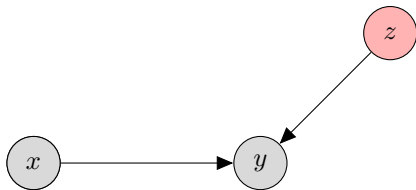


# Controles malos



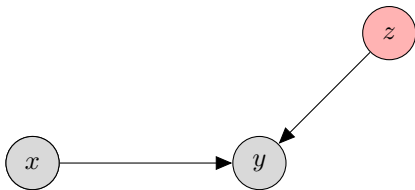
# Controles neutrales

Mejoran precisión



# Controles neutrales

Mejoran precisión

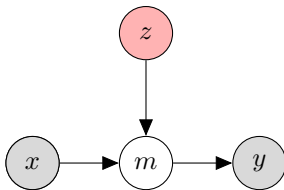


## Efectos causales heterogéneos

Distintos en función de las características de las personas

# Controles neutrales

Mejoran precisión

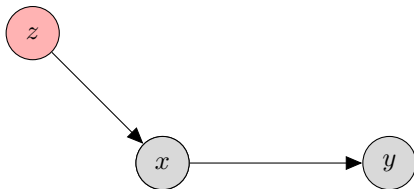


## Efectos causales heterogéneos

Distintos en función de las características de las personas

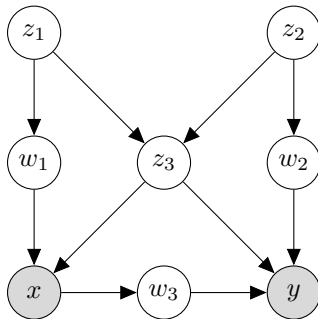
# Controles neutrales

Reducen precisión

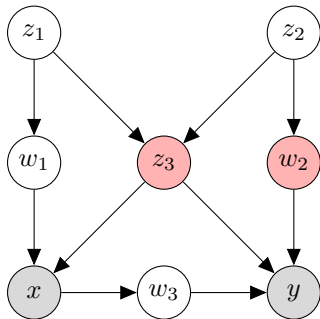


# Estimación de efecto causal

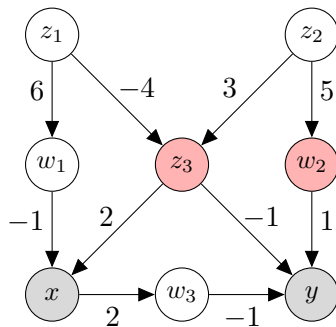
Realidad causal subyacente



# Estimación de efecto causal



# Estimación de efecto causal



# Estimación de efecto causal

## Modelos lineales

```
import numpy as np
```

```
N = 1000
```

```
z1 = np.random.uniform(-3,3, size=N)
```

```
w1 = 3*z1 + np.random.normal(size=N, scale=1)
```

```
z2 = np.random.uniform(-3,3, size=N)
```

```
w2 = 2*z2 + np.random.normal(size=N, scale=1)
```

```
z3 = -2*z1 + 2*z2 + np.random.normal(size=N, scale=1)
```

```
x = -1*w1 + 2*z3 + np.random.normal(size=N, scale=1)
```

```
w3 = 2*x + np.random.normal(size=N, scale=1)
```

```
y = 2 - 1*w3 - z3 + w2 + np.random.normal(size=N, scale=1)
```

# Los 4 pasos de la estimación de efecto causales

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1. Especificar la estructura causal subyacente.

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# Los 4 pasos de la estimación de efecto causales

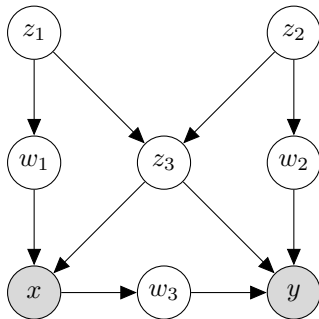
1. Especificar la estructura causal subyacente.
2. Definir el estimando (variables de control y modelo)
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4. Validar los resultados

# Los 4 pasos de la estimación de efecto causales

1. **Especificar la estructura causal subyacente.**
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# Los 4 pasos de la estimación de efecto causales

## Especificar la estructura causal subyacente



# Los 4 pasos de la estimación de efecto causales

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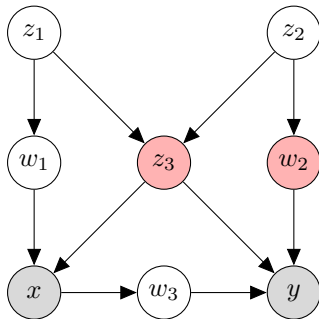
# Los 4 pasos de la estimación de efecto causales

Definir el estimando (variables de control y modelo)

1. **Especificar la estructura causal subyacente.**
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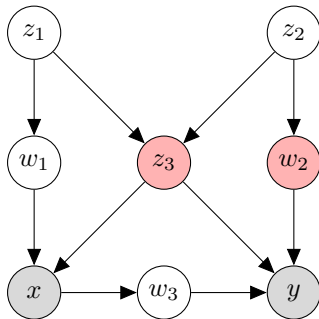
# Los 4 pasos de la estimación de efecto causales

Definir el estimando (variables de control y modelo)



# Los 4 pasos de la estimación de efecto causales

Definir el estimando (variables de control y modelo)



$$y \sim x + z_3 + w_2$$

# Los 4 pasos de la estimación de efecto causales

1. **Especificar la estructura causal subyacente.**
2. **Definir el estimando (variables de control y modelo)**
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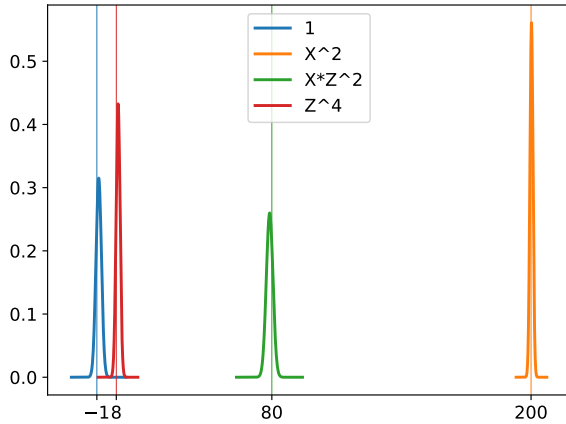
# Los 4 pasos de la estimación de efecto causales

## Computar las estimaciones de impacto causal

1. **Especificar la estructura causal subyacente.**
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# Los 4 pasos de la estimación de efecto causales

## Computar las estimaciones de impacto causal



# Los 4 pasos de la estimación de efecto causales

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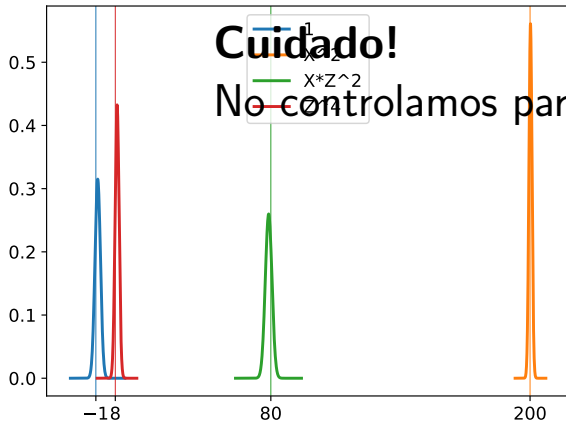
# Los 4 pasos de la estimación de efecto causales

## Validar los resultados

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3. **Computar las estimaciones de impacto causal**
4. **Validar los resultados**

# Los 4 pasos de la estimación de efecto causales

## Validar los resultados



**Cuidado!**

No controlamos para  $z_3$  ni  $w_2$ !

# Estimación de efecto causal

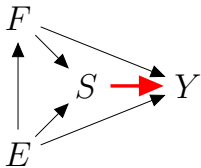
## Table 2 Fallacy

Es una falacia presentar los coeficientes de los controles también como efectos causales

# Estimación de efecto causal

## Table 2 Fallacy

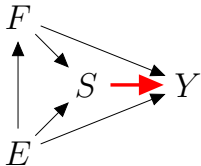
$F$ : Fuma.  
 $E$ : Edad.  
 $S$ : Salud.  
 $Y$ : Síntoma



# Estimación de efecto causal

## Table 2 Fallacy

$F$ : Fuma.  
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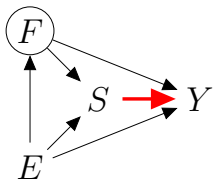


$$S \longleftarrow F \longrightarrow Y$$

# Estimación de efecto causal

## Table 2 Fallacy

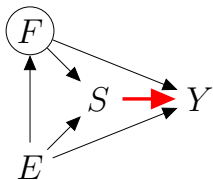
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# Estimación de efecto causal

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# Estimación de efecto causal

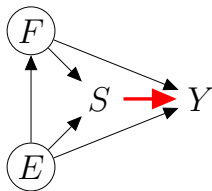
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# Estimación de efecto causal

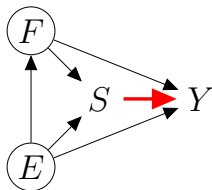
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# Estimación de efecto causal

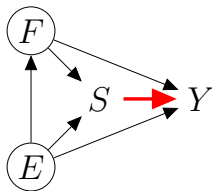
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# Estimación de efecto causal

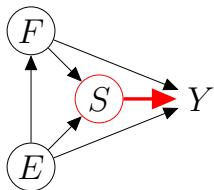
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$F$ : Fuma.

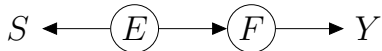
$E$ : Edad.

$S$ : Salud.

$Y$ : Síntoma



$$y \sim c_0 + c_s s + c_f f + c_e e$$



# Estimación de efecto causal

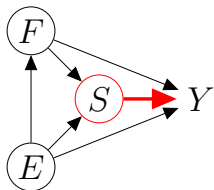
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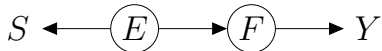
$E$ : Edad.

$S$ : Salud.

$Y$ : Síntoma



$$y \sim c_0 + c_s s + \underbrace{c_f f + c_e e}_{\text{Efectos causales directos}}$$



# Estimación de efecto causal

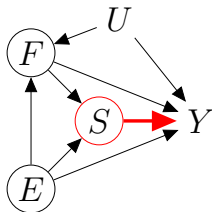
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$F$ : Fuma.

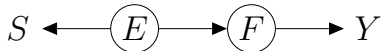
$E$ : Edad.

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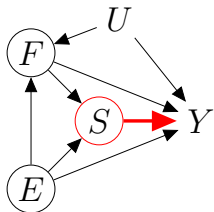
$$y \sim c_0 + c_s s + c_f f + c_e e$$



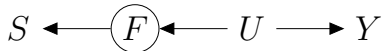
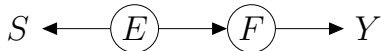
# Estimación de efecto causal

## Table 2 Fallacy

$F$ : Fuma.  
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$$y \sim c_0 + c_s s + c_f f + c_e e$$



# Estimación de efecto causal

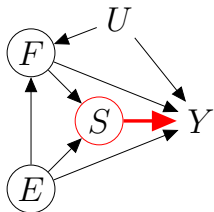
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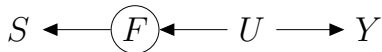
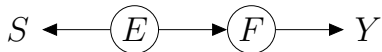
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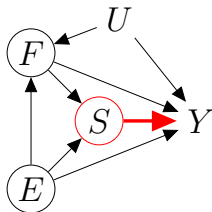
$$y \sim c_0 + c_s s + c_f f + c_e e$$



# Estimación de efecto causal

## Table 2 Fallacy

$F$ : Fuma.  
 $E$ : Edad.  
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 $Y$ : Síntoma



$$y \sim c_0 + c_s s + \underbrace{c_f f + c_e e}_{\text{No son efectos causales!}}$$



# Estimación de efecto causal

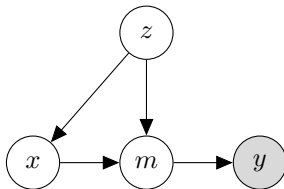
## Modelo polinomial

$$P(z) = \mathcal{N}(z|0, 1)$$

$$P(x|z) = \mathcal{N}(x|z^2, 1)$$

$$P(m|x, z) = \mathcal{N}(m|2z^2 + 10x, 1)$$

$$P(y|m) = \mathcal{N}(y|-1 + 2m^2, 1)$$



# Estimación de efecto causal

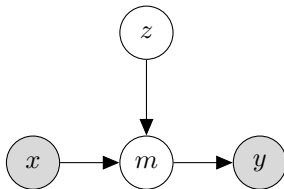
## Modelo polinomial

$$P(z) = \mathcal{N}(z|0, 1)$$

$$P(\text{do}(x)) = \mathbb{I}(X = x)$$

$$P(m|x, z) = \mathcal{N}(m|2z^2 + 10x, 1)$$

$$P(y|m) = \mathcal{N}(y|-1 + 2m^2, 1)$$



# Estimación de efecto causal

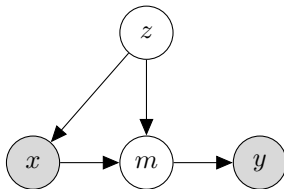
## Modelo polinomial

$$P(z) = \mathcal{N}(z|0, 1)$$

$$P(x|z) = \mathcal{N}(x|z^2, 1)$$

$$P(m|x, z) = \mathcal{N}(m|2z^2 + 10x, 1)$$

$$P(y|m) = \mathcal{N}(y|-1 + 2m^2, 1)$$



# Estimación de efecto causal

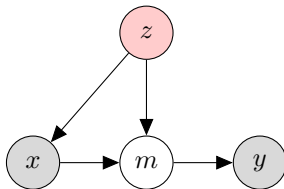
## Modelo polinomial

$$P(z) = \mathcal{N}(z|0, 1)$$

$$P(x|z) = \mathcal{N}(x|z^2, 1)$$

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$$P(y|m) = \mathcal{N}(y|-1 + 2m^2, 1)$$



# Estimación de efecto causal

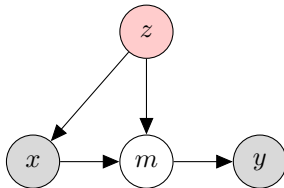
## Modelo polinomial

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$$P(y|\text{do}(x), z) = P(y|x, z) = \mathcal{N}(y|\mathbb{E}[y|x, z], \mathbb{V}[y|x, z])$$

# Estimación de efecto causal

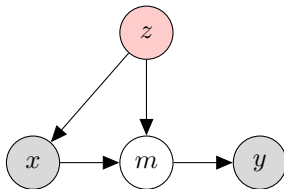
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¿Qué polinomio debemos usar para modelar la regresión lineal?

# Estimación de efecto causal

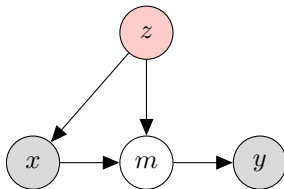
## Modelo polinomial

$$P(z) = \mathcal{N}(z|0, 1)$$

$$P(x|z) = \mathcal{N}(x|z^2, 1)$$

$$P(m|x, z) = \mathcal{N}(m|2z^2 + 10x, 1)$$

$$P(y|m) = \mathcal{N}(y|-1 + 2m^2, 1)$$



$$P(y|\text{do}(x), z) = P(y|x, z) = \mathcal{N}(y|\mathbb{E}[y|x, z], \mathbb{V}[y|x, z])$$

Siempre podemos evaluar modelos alternativos en base a la evidencia

# Estimación de efecto causal

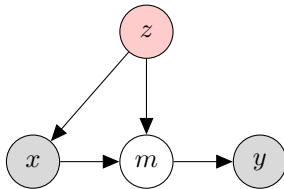
## Modelo polinomial

$$P(z) = \mathcal{N}(z|0, 1)$$

$$P(x|z) = \mathcal{N}(x|z^2, 1)$$

$$P(m|x, z) = \mathcal{N}(m|2z^2 + 10x, 1)$$

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$$P(y|\text{do}(x), z) = P(y|x, z) = \mathcal{N}(y|\mathbb{E}[y|x, z], \mathbb{V}[y|x, z])$$

Pero acá conocemos la función.

# Estimación de efecto causal

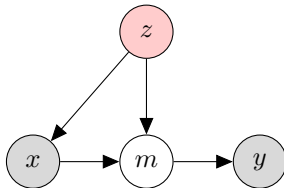
## Modelo polinomial

$$P(z) = \mathcal{N}(z|0, 1)$$

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$$P(y|\text{do}(x), z) = P(y|x, z) = \mathcal{N}(y|\mathbb{E}[y|x, z], \mathbb{V}[y|x, z])$$

$$\mathbb{E}[y|m] = -1 + 2m^2$$

# Estimación de efecto causal

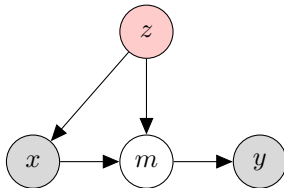
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$$P(y|\text{do}(x), z) = P(y|x, z) = \mathcal{N}(y|\mathbb{E}[y|x, z], \mathbb{V}[y|x, z])$$

$$\mathbb{E}[y|m] = -1 + 2m^2 = -1 + 2(2z^2 + 10x)^2$$

# Estimación de efecto causal

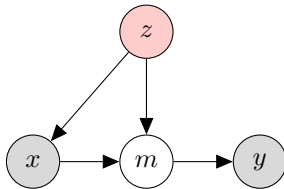
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$$P(m|x, z) = \mathcal{N}(m|2z^2 + 10x, 1)$$

$$P(y|m) = \mathcal{N}(y|-1 + 2m^2, 1)$$



$$P(y|\text{do}(x), z) = P(y|x, z) = \mathcal{N}(y|\mathbb{E}[y|x, z], \mathbb{V}[y|x, z])$$

$$\begin{aligned}\mathbb{E}[y|m] &= -1 + 2m^2 = -1 + 2(2z^2 + 10x)^2 \\ &= -1 + 200x^2 + 80xz^2 + 8z^4\end{aligned}$$

# Estimación de efecto causal

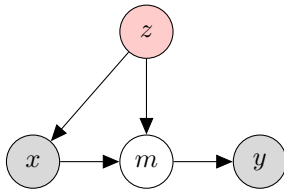
## Modelo polinomial

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$$P(y|m) = \mathcal{N}(y|-1 + 2m^2, 1)$$



$$P(y|\text{do}(x), z) = P(y|x, z) = \mathcal{N}(y|\mathbb{E}[y|x, z], \mathbb{V}[y|x, z])$$

$$\mathbb{E}[y|m] = -1 + 2m^2 = -1 + 2(2z^2 + 10x)^2$$

$$= -1 + 200x^2 + 80xz^2 + 8z^4 = \mathbb{E}[y|x, z]$$

# Estimación de efecto causal

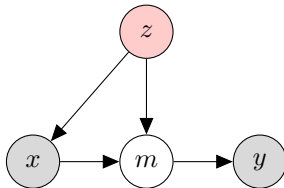
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$$\mathbb{E}[y|x, z] = -1 + 200x^2 + 80xz^2 + 8z^4$$

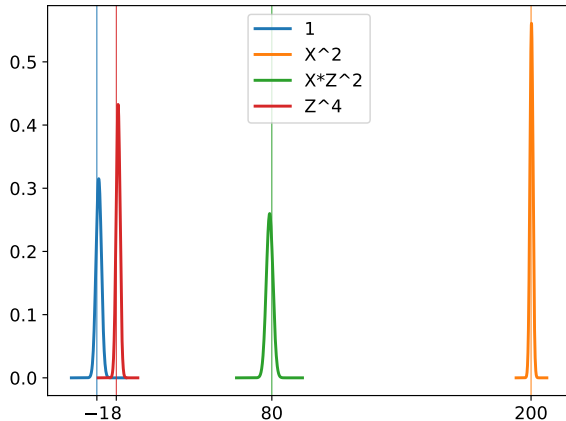
# Estimación de efecto causal

## Modelo polinomial

```
Z4s = np.random.normal(size=N, scale=1)
X4s = Z4s**2 + np.random.normal(size=N,scale=1)
M4s = 2*Z4s**2 + 10*X4s + np.random.normal(size=N,scale=1)
Y4s = -1 + 2*M4s**2 + np.random.normal(size=N,scale=1)
```

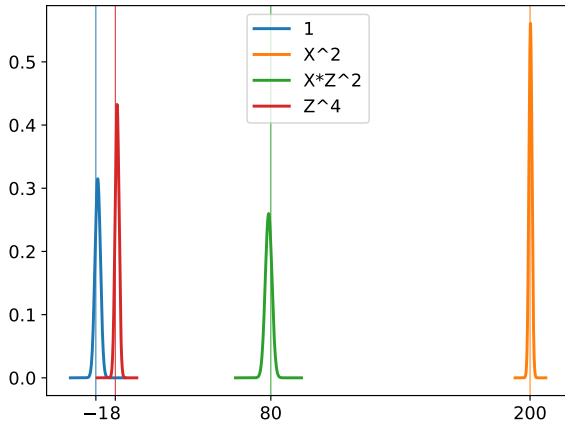
# Estimación de efecto causal

## Modelo polinomial



# Estimación de efecto causal

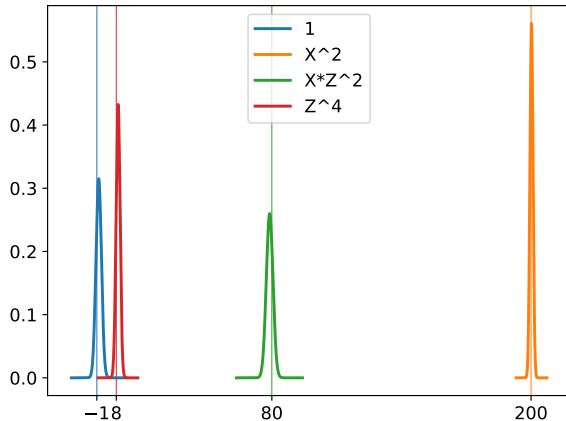
## Modelo polinomial



$$\mathbb{E}[y|x, z] = -1 + 200x^2 + 80xz^2 + 8z^4$$

# Estimación de efecto causal

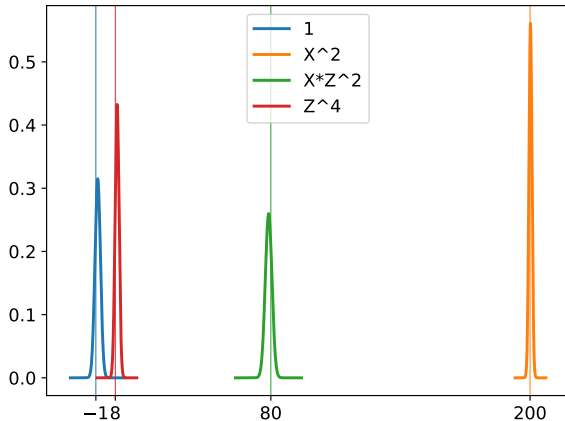
## Modelo polinomial



$$\mathbb{E}[y|x, z = \pm 1] = 7 + 200x^2 + 80x$$

# Estimación de efecto causal

## Modelo polinomial

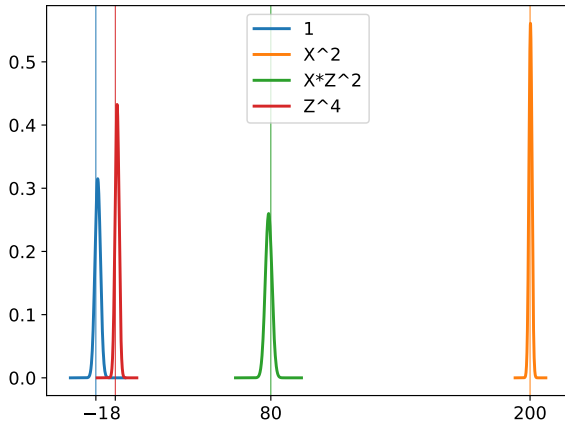


$$\mathbb{E}[y|x, z = \pm 1] = 7 + 200x^2 + 80x$$

$$\mathbb{E}[y|x, z = \pm 2] = 47 + 200x^2 + 320x$$

# Estimación de efecto causal

## Modelo polinomial



¿Y si queremos saber el efecto promedio  $P(y|\text{do}(x))$ ?

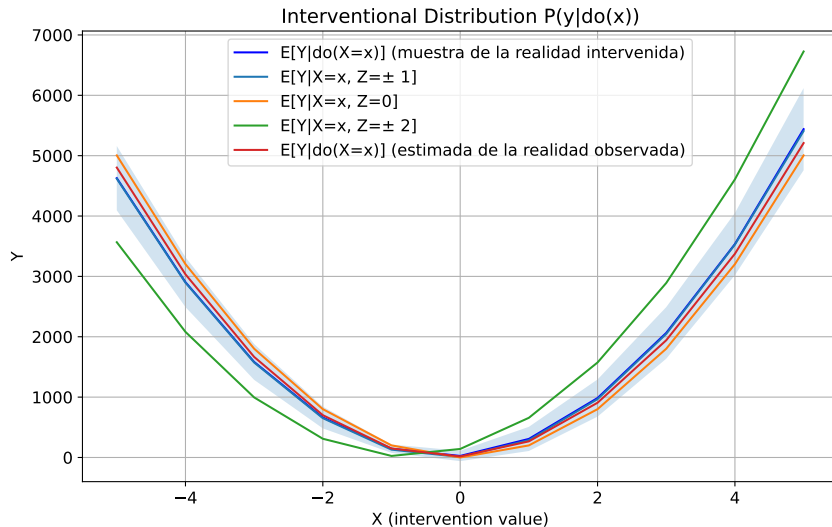
# Estimación de efecto causal promedio

## Modelo polinomial

```
def p4_Y_doX(y,x, beta=beta4):  
    res = 0  
    z_grilla = np.linspace(-4,4,201)  
    dz = z_grilla[1]-z_grilla[0]  
    for z in z_grilla:  
        pz = normal(0,1).pdf(z)  
        new_Phi= np.array([1,x**2,x*z**2,z**4]).reshape((1,4))  
        mu, sigma2 = moments_predictive(new_Phi, beta = beta4,  
            t_priori=Y4s, Phi_priori=PHI4)  
        py_xz = norm(loc=mu, scale=np.sqrt(sigma2)).pdf(y)  
        res += (pz*py_xz)*dz  
    return res
```

# Estimación de efecto causal promedio

## Modelo polinomial



$p = \mathbf{b}$

Laboratorios de  
Métodos Bayesianos